



Recent Trends in Wind Energy Conversion System with Grid Integration Based on Soft Computing Methods: Comprehensive Review, Comparisons and Insights

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Abstract

Wind energy is an effective and promising renewable energy source to produce electrical energy. Wind energy conversion systems (WECS) have been developing on a wide scale worldwide. The expansion of wind energy demand tends to produce high-quality output power in terms of grid integration. Due to the intermittent nature of wind energy, great challenges are found regarding WECS modeling, control, and grid integration. This paper introduces a comprehensive review of WECS and their grid-interface systems based on soft computing methods. To achieve this aim, more than 300 articles are organised and only 160 papers are presented in this review. This is intended to cover a broad range of topics concerning the configurations of WECS, electrical generators, and various topologies of power converters used for control and grid integration. Furthermore, international grid codes for wind energy integration with electric grids, particularly frequency, power factor, and low voltage ride through (LVRT) capability are investigated. The major controller approaches and topologies for grid and generator converters are discussed. Different aspects of modern control of WECS are introduced either for grid-side or generator-side. Moreover, control strategies for maximum power point tracking methods are compared along with methods of frequency control. This review paper introduces a comprehensive and a useful summary for the recent work in literature regarding WECS. Detailed modelling, control, and grid integration along with comparisons and discussion are introduced.

1 Introduction

The change from fossil fuels to sustainable energy resources is becoming an international obligation. The main causes for this shift are harmful impacts on the environment, particularly climate and the lack of fossil fuels. Therefore, the renewable energy sources like solar, wind, wave energy are getting a lot of attention from the world [1]. With low demand for all other fuels, the use of renewable energy increased 3% in 2020. While in 2021, the generation of electricity from renewable energy is set to expand more than 8% to 8300 TWh, the fastest annual growth rate since the 1970s [2]. The wind and solar PV energy are expected to generate two-thirds of renewable energy growth. In 2021, China alone produces nearly half of the worldwide growth in renewable energy followed by the USA, European Union

and India. The increase rate of renewable electricity generation for the years 2019–2020 and 2020–2021 based on TWh values and year-on-year growth is shown in Fig. 1a. The high efficiency and the relatively low cost of wind power generation are the reasons of why wind energy is the largest compared to other renewable energy resources [3]. The production of electricity from wind energy is rapidly developing among many renewable energy sources as shown in Fig. 1a. The installed wind capacity of the planet reached 175 TWh during 2019–2020, while it increases during 2020–2021 to 275 TWh [2].

A wind energy conversion system (WECS) is used to produce electrical energy from wind in a reliable, controlled, and efficient way. Figure 1b shows the layout of a WECS. The main components of a WECS can be generally categorised into electrical, mechanical and control systems. For the electrical components, they consist of electrical generator, possible grid-side and generator-side converters, step-up transformer, harmonic filters and three-phase grid. The mechanical components include the nacelle, etc. The control related components are employed with both the electrical and mechanical energy conversion

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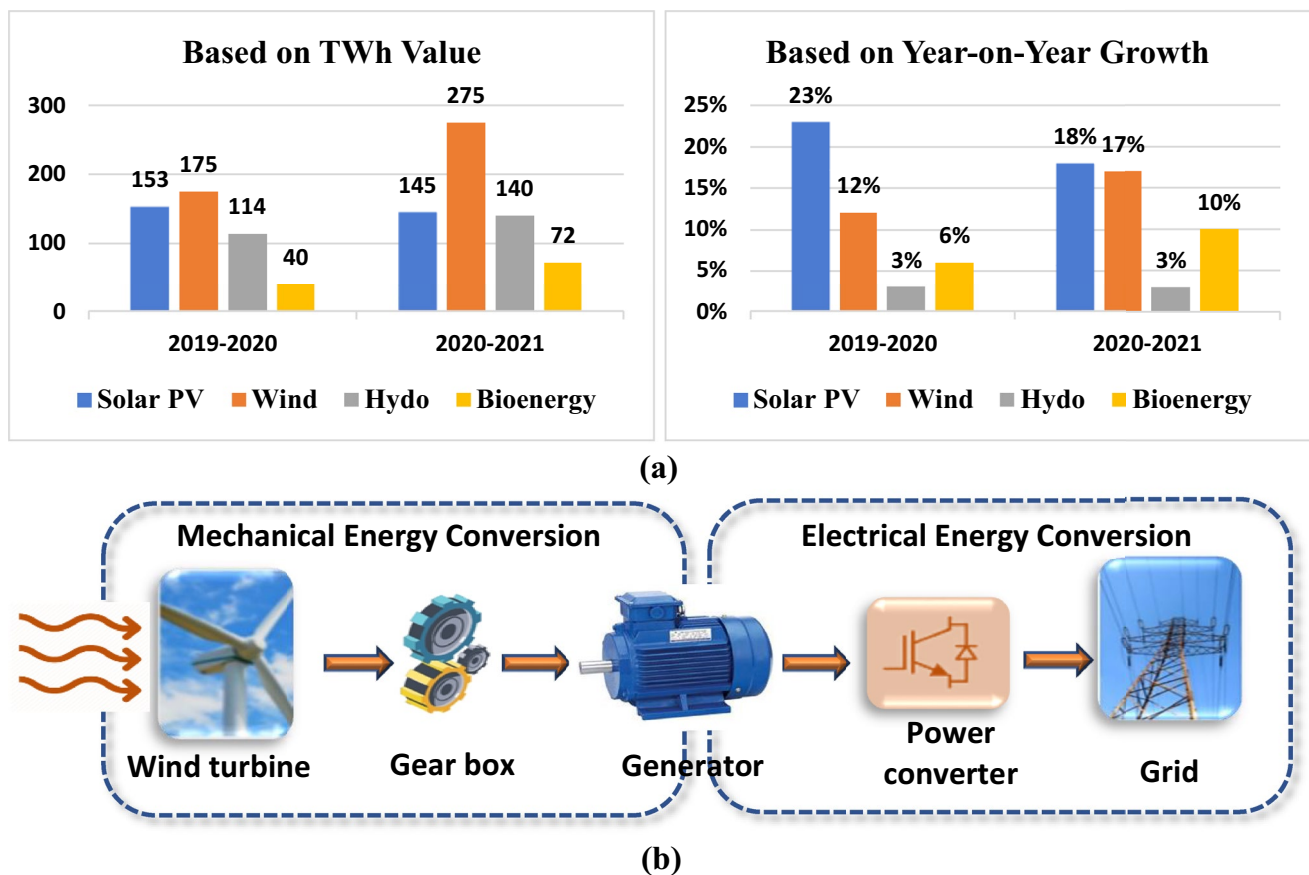


Fig. 1 a The increase rate of renewable electricity generation based on TWh values and year-on-year growth, b Schematic diagram of a WECS

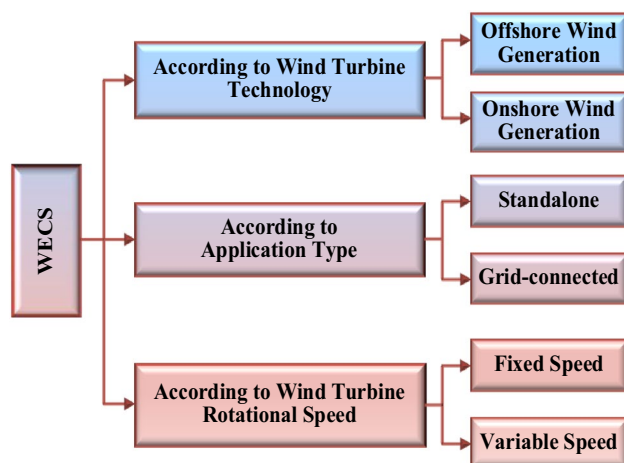


Fig. 2 The classification of WECSs

systems [4]. The classification of WECSs is shown in Fig. 2. With regard to the technology of wind turbines, there are two main types which are offshore and onshore wind generation. Because of the low cost of maintenance and installation of onshore wind farms by comparison

with offshore wind farms, onshore generation represents nearly 96% of the total global wind energy [5]. According to the application type, WECS is mainly classified into two categories, standalone and grid-connected WECS. For standalone WECS, it is a substitute option for power consumers in locations where the expansion of the electrical grid is too costly. Because of changing load demand with regard to time and the output power of turbine with the variation of wind speed, standalone WECS is used in integration with other types of power generation [6]. For grid-connected WECS, the connection between the grid and the turbine is carried out at different levels of voltage. The uncontrolled and variation of wind generation is one of the challenges that makes connecting to the grid a difficult task for the operators of system to make the power system balanced and stable. Additionally, the high penetration of wind generation affects various aspects of the power system operation like stability power quality, and economics [7]. According to the rotational speed of wind turbine, WECS is principally classified into two categories, variable speed and fixed speed. Fixed speed systems have led the way owing to their cheap initial costs, simplicity, and high reliability. They were prevalent until around two

decades ago, when variable speed systems began to take their place. This is owing to its intrinsic drawbacks, which include inflexibility in supporting grid voltage adjustment, mechanical stress difficulties, low wind energy conversion efficiency, and inevitable power flicker. Moreover, any variation in wind power has an immediate impact on the grid [8]. Variable speed systems are attracting more popularity. They have several advantages, including rapid response under transient power system conditions, increased wind energy conversion efficiency and enhanced power quality. Furthermore, by employing a high-pole number electric machine, the requirement for a gearbox may be removed. Higher losses from the additional devices of power electronic, and also increased cost and the complexity of control, are some of its drawbacks.

The distribution and transmission system operators face several problems as a result of the integration of large-capacity wind power plants into the grid, including electrical power system reliability and stability and power quality. As a result, a number of specific technical requirements known as grid codes have been created and are being updated on a regular basis [9]. The most prevalent grid code requirements are fault ride through (FRT) capabilities, as well as frequency and voltage change limitations, the control of voltage, reactive power, active power and frequency [10]. Owing to the rapid advancement of power electronic technology over the last three decades, power electronics have played an essential role in the integration of large-scale wind systems into the grid. In recent wind power generation plants, power converters are essential in order to provide independent active and reactive power control, variable speed operation and uncomplicated maximum power point tracking (MPPT) operations [11].

The contribution of this paper can be summarized as: (i) An intensive overview about grid-connected WECSs, including a review on electrical generators and power converters, (ii) An insight on different requirements of grid codes and various controllers that are used in wind generation systems, (iii) Recent approaches of LVRT, MPPT and frequency control and (iv) Artificial intelligence (AI) techniques that are used in grid-connected WECSs. This paper is organised as follows: Sect. 2 presents a review of the electrical machines for wind power generation. In Sect. 3, a review of the power converters which are used in wind power generation is introduced. Section 4 describes the different configurations of grid-connected wind systems. Section 5 presented an overview about standard grid codes for wind power integration around the world. Section 6 presents an overall view of the electrical controllers that are used for WECSs. Section 7 presented an overall view of AI techniques that are used in grid-connected WECSs. Finally, the conclusion is presented at the end of this paper.

2 Electrical Generators for WECSs

This section will cover an overview of the electrical machines for wind power generation. Currently, DC generators are often used in low voltage and low power standalone systems therefore, they are limited to very small wind applications. As for the modern trend, it is directed towards AC generators which are the most popular option for large-scale wind applications. So, the major types of electrical machines which are used for wind power generation are induction and synchronous generators as shown in Fig. 3 [12]. One of the negative sides of using induction generators in WECS is that a gear system is required [13, 14]. There are two types of synchronous generators utilised in the WECS: cylindrical pole and salient pole [15, 16]. The elimination of gear system is one of the benefits of synchronous generators over induction generators. The removal of the gear system decreases costs and losses and improves efficiency. It is necessary to increase the number of poles in the rotor to achieve the required turbine speed [17].

Squirrel cage induction generator (SCIG) is considered one of the first machines employed in wind energy systems. SCIGs are usually used in fixed/semi variable speed systems. Complicated power factor (PF) improvement control schemes are needed because of the consumption of reactive power from grid by SCIGs to maintain their operation of generation. With regard to the advantages of SCIGs, they are commonly accepted choices for WECS where, they have high reliability, low cost and the operation with no maintenance. However, the major SCIGs drawbacks are large starting current and low starting torque [15, 18–20].

Wound rotor induction generators (WRIG) are also among the first machines used in WECS. For WRIGs, external resistance is connected to the windings of rotor via slip rings. By regulating the external resistance, WRIGs have large starting torque and small inrush current. Additionally,

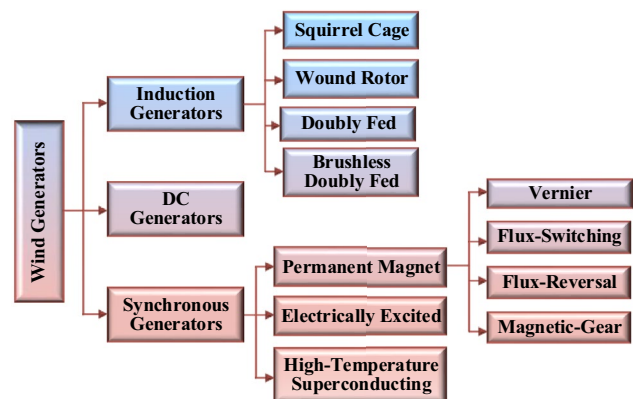


Fig. 3 The various categories of electrical machines used in a WECS

PF can be enhanced by WRIGs in comparison to SCIGs. However, the disadvantages related to WRIGs are complexity and the requirement for periodic maintenance for slip rings and brushes. WRIGs are generally used in fixed/semi variable speed systems. Power factor improvement is essential because WRIGs consume reactive power from the grid [15, 21, 22].

Doubly fed induction generators (DFIG) are an extension of the evolution of WRIGs where, they become more widespread in the generation of wind energy. DFIG is a WRIG with a wound rotor with a slip ring and brushes connected to controller, in place of external resistance. For compensating the fluctuations of turbine speed, the rotor input current is adapted in phase angle and frequency. Furthermore, reactive and active power injected to the grid can also be optimized by this adaption of the rotor input current. Therefore, DFIGs are convenient for variable speed wind systems [15, 23, 24]. Brushless doubly fed induction generator (BDFIG) has attracted wind attention because of the avoidance of the problems related with slip rings and brushes. Many versions of BDFIG have been evolved from traditional cascaded generators to self-cascaded generators and to modern DFIGs [25]. Because of the absence of brushes and slip rings, the advantages of BDFIGs are low maintenance and the increasing of reliability, besides the ease of reactive and active power control makes BDFIGs promising prospects in variable speed WECS. The grid is connected to the power windings, whereas the control windings are fed through a controller and converter in back-to-back configuration in order to flow the power in two directions [26]. The research on BDFIGs is still in the preliminary phase. The reason for this is the difficulty of implementing BDFIGs in practice because of power control intricacy [15, 24, 27].

For electrically excited synchronous generator (EESG), DC excitation may be either brushless exciters or an external DC source supplied through slip rings and brushes. In order to fulfil the turbine speed requirement, the number of poles must be raised [17, 28, 29]. In the year 1980, high-temperature superconducting (HTS) material was developed. The field winding for the high-temperature superconducting synchronous generator (HTS-SG) is made of a super conductivity material like Bi-2223 or BSCCO. This material operates at a high temperature (35–40 K). An off-shelf cooling system is used to easily cool this material [17]. The machine armature is designed in such a way that the density of magnetic flux in the stator is maximized, allowing the line voltage to remain constant [17, 30–32].

Permanent magnet synchronous generator (PMSG) is the most widely used generator among synchronous generators in WECS. There are two styles of PMSG: inset-mounted and surface-mounted PMSG. For low-speed stand-alone applications, the surface-mounted PMSG is employed. While, the inset mounted PMSG is more suited for high-speed

applications [16, 26, 27]. There are some modifications to PMSGs to improve their performance in wind systems where these generators are Vernier, flux-switching, flux-reversal, and magnetic-gear PMSGs.

Vernier PMSG generators are gaining popularity because of their simple mechanical construction and large torque density. The topology of Vernier PMSG is the same structure of a traditional surface-mounted PMSG. For conventional PMSGs, the number of rotor and stator poles is equal, while Vernier PMSGs have a different number for rotor and stator poles. In general, Vernier PMSGs have a fairly small stator poles number and a greatly high rotor poles number to emphasise the effect of Vernier. The torque of Vernier PMSG increases when the rotor speed is less than the rotating field speed. Therefore, they are suitable choice for applications that require high-torque and low-speed such as wind power generation [17, 33–37]. Flux-switching PMSG (FS-PMSG) has permanent magnets on the stator unlike the traditional PMSGs which have permanent magnets on the rotor. Because of the simplicity of construction, the rotor of FS-PMSGs is mechanically durable. In direct-drive WECSs, a high number of poles are necessary to remove gearboxes. Because of the merits of FS-PMSGs such as basically sinusoidal back-EMF, the high density of torque, good thermal management of permanent magnets, and they have gained a lot of attention in the previous few decades [17, 38, 39]. Flux-reversal PMSGs (FR-PMSG) have attracted attention in recent decades due to their benefits; fast transient response, simplicity of rotor configuration, and high-power density. Moreover the mechanical robustness of FR-PMSG rotor. Each stator tooth has a pair of permanent magnets with alternating polarities placed on the inside surface. In addition, the windings are commonly coiled around the teeth of stator, causing short-end windings. Back-EMF waveforms in the tooth windings are generated by bipolar flux [17, 40]. The concept of magnetic-gear PMSGs (MG-PMSG) is magnetic gear. The mechanism of operation is based on magnetic field modulation generated by two permanent magnet rotors by using ferromagnetic pole pieces [17]. The permanent magnets on the rotor and the armature windings in the stator are designed to have various pole numbers, resulting in varying speeds. The rotating ferromagnetic pole pieces act as a flux modulator, ensuring that the stator and rotor magnetic fields have the same speed and pole number. As a result, steady torque is produced, similar to what a reduction gear does, resulting in low speed and high torque which is suitable in WECSs [17, 41, 42].

Table 1 summarises the major differences between common electrical generators that are used in WECSs. According to surveys, the industry uses DFIGs by approximately 48.6%, SCIGs by 48.1%, PMSGs by 3.2%, and the remaining machine types are around 0.1% [15]. Therefore, induction generators dominate the wind market with a large percentage

Table 1 The major differences between the common electrical wind generators

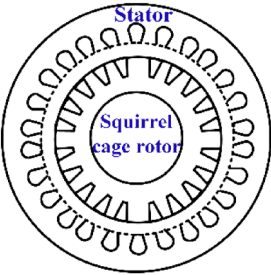
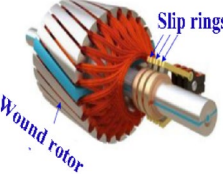
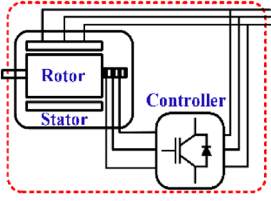
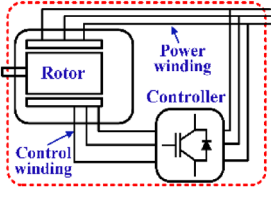
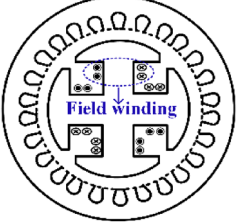
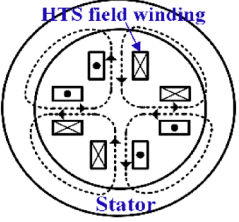
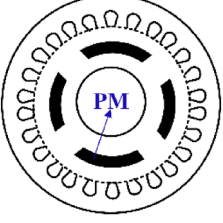
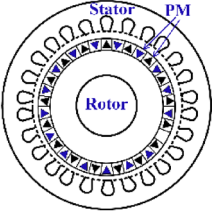
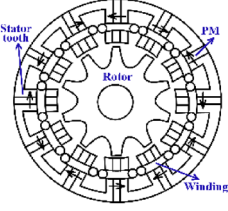
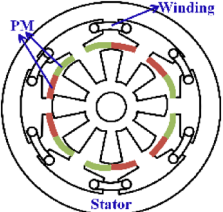
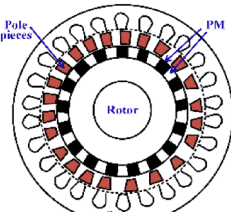
	Construction	Speed range	Gear box	Soft starter	Power range	Efficiency	Cost
SCIGs		Fixed/semi variable speed (< 1%)	Multi-stage gear	Required	Low	Low	Low
WRIGs		Fixed/semi variable speed (< 10%)	Multi-stage gear	Required	Low	Low	Low
DFIGs		Variable SPEED ($\pm 30\%$)	Multi-stage gear	Not required	High	High	High
BDFIGs		Variable speed (0–100%)	Multi-stage gear	Not required	High	High	High
EESGs		Variable speed (0–100%)	No	Not required	Medium	High	Medium
HTS-SGs		Variable speed (0–100%)	No	Not required	High	Very high	Medium
PMSGs		Variable speed (0–100%)	Optional	Optional	Low to medium	High	Medium

Table 1 (continued)

	Construction	Speed range	Gear box	Soft starter	Power range	Efficiency	Cost
Vernier PMSGs		Variable speed (0–100%)	No	Not required	Low	Low	High
FS-PMSGs		Variable speed (0–100%)	Optional	Not required	Both low and high	Low	Low
FR-PMSGs		Variable speed (0–100%)	No	Not required	Low	Low	Low
MG-PMSGs		Variable speed (0–100%)	No	Not required	Low	Low	Low

of about 97%. The most widely used of the induction machines is BDFIG thanks to the avoidance of brushes, slip rings, and losses in rotor windings. The induction machines are not suitable for gearless drives due to the losses of excitation in these large machines as a result of the large air gap. The possibility of removing the gear system in the wind generator is one advantage of using the synchronous generator. The main problems of using PMSGs are the inability to regulate their excitation and the high cost of magnets. Nowadays, the cost of permanent magnets is falling substantially.

2.1 Recent Trend in Wind Generators

This subsection introduces the overall view of the recent research on the enhancement of the performance of wind generators. In [43], Liu et al. introduced a unique bearing current reduction technique by using the electrostatic shield on the DFIGs rotor side to mitigate the ratio of bearing voltage in the generator. This research proved that the bearing currents can be efficiently reduced by the novel suppression

approach. Under the unsegmented state, the suppression effect is better, but the eddy current loss is greater. In [44], Mishra et al. presented a novel six-phase DFIG with asymmetrical winding structure to capture electric power from WECS. The attractive benefits of the proposed novel machine above the conventional induction machines are high reliability and efficiency, reduced per leg converter rating, low pulsating torque and per phase current, and the possibility of using for feeding 6-phase and 3-phase transmission line. In [45], Sourabh and Parida presented a novel design of BDFIG that considers a U-shaped rotor and dual-stator structure. The major advantages of this type of design are high power density and torque with low torque ripple, wide range of speed and torque in stable region, operating at synchronous speed like synchronous machines. In [46], Zhang et al. introduced a novel interior permanent magnet BDFIG for wind power applications, that is based on the magnetic field modulation theory. The merits of this design are high torque density and power density, and high permanent magnet utilisation. In [47], Zhang et al. presented a novel BDFIG

with the hybrid rotor that is used in large-scale offshore wind power generation with number of distinct benefits. In [48], Zhang et al. presented BDFIG with novel hybrid rotor with a new technique based on the magnetic field modulation theory for the best selection of slot combination.

In [49], Cheng et al. introduced a novel 10 MW HTS synchronous machine with extra permanent magnets installed between the pole-tips of silent rotor for offshore wind energy applications. This study concluded that the proposed generator offers advantages in terms of torque density increase and mass reduction, as well as general short-circuit fault performance in comparison to traditional machines. In [50], Dong et al. presented a novel radial partitioned stator HTS-excitation flux-switching machine. The key is that the single ring-shaped HTS-excitation coil is independently placed in the coaxial claw-pole outer stator. While the armature windings are wound around the U-shaped iron core on the inner flux-switching armature stator without permanent magnets. In [31], Zhou et al. introduced a 2 MW direct drive HTS wind generator by using HTS wires in copper transposed conductor in the stator windings and the rotor field coils. In [32], Xue et al. discussed the performance of machines with HTS armature windings which are more sensitive to the parameters of design owing to the interaction between the flux density and the operating current compared with the conventional electrical machines.

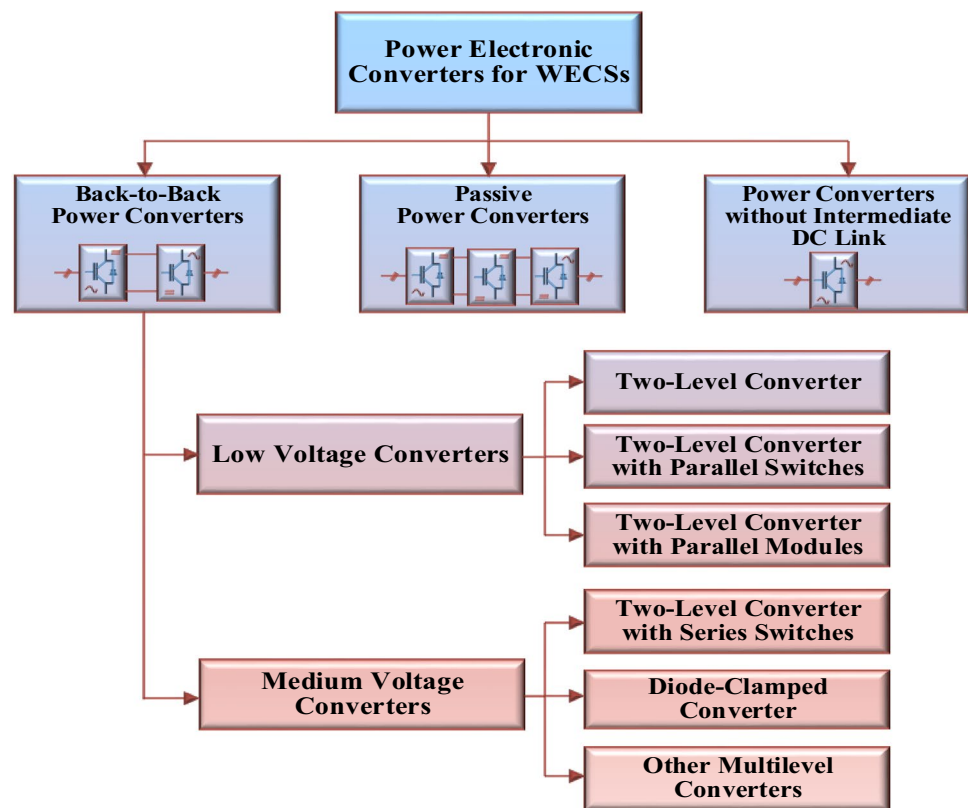
In [51], Kumar et al. introduced the characteristics and design for a novel PMSG for the application of wind power. The rotor of the proposed generator features is a V-shaped embedded magnetic pole and the stator includes six-phase windings. The conclusion of this paper was higher reliability and power density of the proposed generator as compared to the conventional generators. In [52], Yang et al. introduced a novel asymmetric-magnetic-pole interior PMSG by combining the permanent magnet structures of flat-type and dual-layer reluctance rotor. The major merits of this proposed machine are greater high efficiency region because of the improvement of torque, and higher power and torque in the low-speed region. In [53], Xiao et al. proposed a novel topology of spoke-type asymmetric interior PMSG featured by asymmetric flux barrier with 12 slots, 8 poles, and no overlapping stator windings. In [54], Jiang et al. presented a novel doubly-fed doubly-salient machine with DC saturation-relieving effect which can capture more wind power and extend speed range. The proposed machine is suitable for low and high wind power applications. In [55], Zeinali and Keysan introduced a novel electrically excited claw pole Vernier generator for wind turbines. This novel topology has the potential to decrease the bulk and cost of generator, as well as making manufacture and handling easier. In [56], Allahyari and Torkaman presented a novel construction of consequent pole dual rotor Vernier PMSG. Windings are put on the dual-sided stator slots, and magnets are

positioned on the inner and outer rotor slots in this structure. In [57], Jia et al. introduced a novel stator and rotor dual permanent magnet Vernier machine with multiple magnetic sources. Genetic algorithm and finite element method were employed to optimize the number of slots and poles where the optimal numbers are 12 slots and 10 poles. In [58], Siddiqi et al. introduced a novel topology of dual air gap radial flux Vernier PMSG with. The proposed machine consists of two stators and a sandwiched yokeless rotor that aids in the reduction of magnet volume by allowing for effective flux coupling in the stator windings. In [59], Chen et al. introduced a novel FS-PMSG with dual sets of magnet arrangements. Compared with the conventional multi-tooth FSPM machine, it is demonstrated that the proposed topology can achieve 79.4% lower cogging torque, 30.8% higher torque density, and 15.6% higher power factor than the traditional machine. In [60], Jiang et al. presented a novel 24/17-pole 3-phase stator wound field FS-PMSG with the combination of asymmetric stator poles and overlapping armature winding for higher torque density. In [61], Yang et al. proposed a novel FR-PMSG with asymmetric stator pole configuration. Unlike traditional FRPM machines, which have regular “NS-NS-NS” permanent magnet sequence, the proposed machine has “NSN-S-NSN” permanent magnet configuration. In [62], Zhou et al. presented a novel field-modulated MG-PMSG. The key design feature of the proposed machine was the use of an inner rotor configuration and a stator core with fixed modulator parts to increase fabrication feasibility and rotation dependability.

3 Power Electronic Converters for WECSs

The connecting of wind turbine to the grid is a vital aspect of the wind-based power system. There are two options for dealing with this issue, one is to employ an appropriate energy storage system, which is difficult to implement owing to the high cost and large space. Another option is the usage of power electronic converters majorly on the grid side with a suitable control mechanism. For variable speed WECSs, by employing the power converter on the generator side, the extraction of maximum power from the wind is possible while keeping the system stable. In addition, by employing the power converter on the grid side, the wind turbine can control the exchange of reactive power and enable it to participate in the regulation of voltage. The power converters must meet certain requirements like limited weight and physical size, reliability, low power losses, and minimum maintenance to be utilized in the WECSs. Many topologies of power converters which take into account these properties have been presented by researchers of WECSs. As illustrated in Fig. 4, the converter topologies are divided into three categories [4, 63, 64].

Fig. 4 Power electronic converters classification for WECSs



3.1 Back-to-Back Power Converters

Back-to-back power converters are two similar converters which are connected back-to-back at the DC link through inductors or capacitors. They convert the variable frequency and voltage of the generator to DC voltage and then to AC with a constant voltage and frequency at the grid side. Back-to-back converters provide a four-quadrant operation for wind generators. Figure 4 shows the various back-to-back power converters that can be utilized in variable-speed WECSs. They are generally divided into low voltage and medium voltage converters [4, 63, 64]. Two-level back-to-back converter is commonly used in WECSs. The generator-side converter is realized by two-level rectifier while, the two-level inverter is employed on grid side as illustrated in Fig. 4a. To obtain voltage and capacitance levels, the DC link is accomplished using a parallel/string of electrolytic capacitors. Furthermore, the DC link allows for decoupling between the grid and generator. As a result, the generator transients do not appear on the grid side. To increase the power ratings, more three-phase two-level converter modules are connected in parallel with common DC-link element for all converters to decrease space and cost. This configuration provides redundancy and energy efficiency. A mismatch in the parameters of grid-side filter and converter causes circulating currents in both grid-side and generator-side converters. This problem must be considered in the

controller design. To solve the reliability and circulating currents issues, the individual DC-link element in each converter module is employed in order not to lose the best qualities like modularity, power handling capability, efficiency, and redundancy. But, the cost of system increases due to the individual DC link [4, 63, 64]. The increasing of power level in wind turbines is driving the technology of power electronics toward medium voltage operation. The medium voltage power converters will dominate the future generation of wind turbines due to their cost-effective, compact, and reliable design. The two-level back-to-back converter shown in Fig. 5a can also be used for medium voltage applications by connecting switching devices in series. To provide same voltages among the series switches, an appropriate voltage equalization method must be developed. Three-level back-to-back neutral-point clamped converter is an alternative solution for medium voltage operation as shown in Fig. 5b. Neutral-point clamped converters lead to low output current ripple, reduced switching losses, and small harmonic filter than two-level back-to-back converters. The voltages across the capacitors should be balanced in order not to damage the semiconductor switches due to high stress on switches [65]. The multilevel converters are promise for future wind turbines because of their technical advantages and suitability for medium voltage operation. The traditional multilevel converter topologies like cascaded H-bridge converter, flying capacitor converter, and hybrid converter provide high

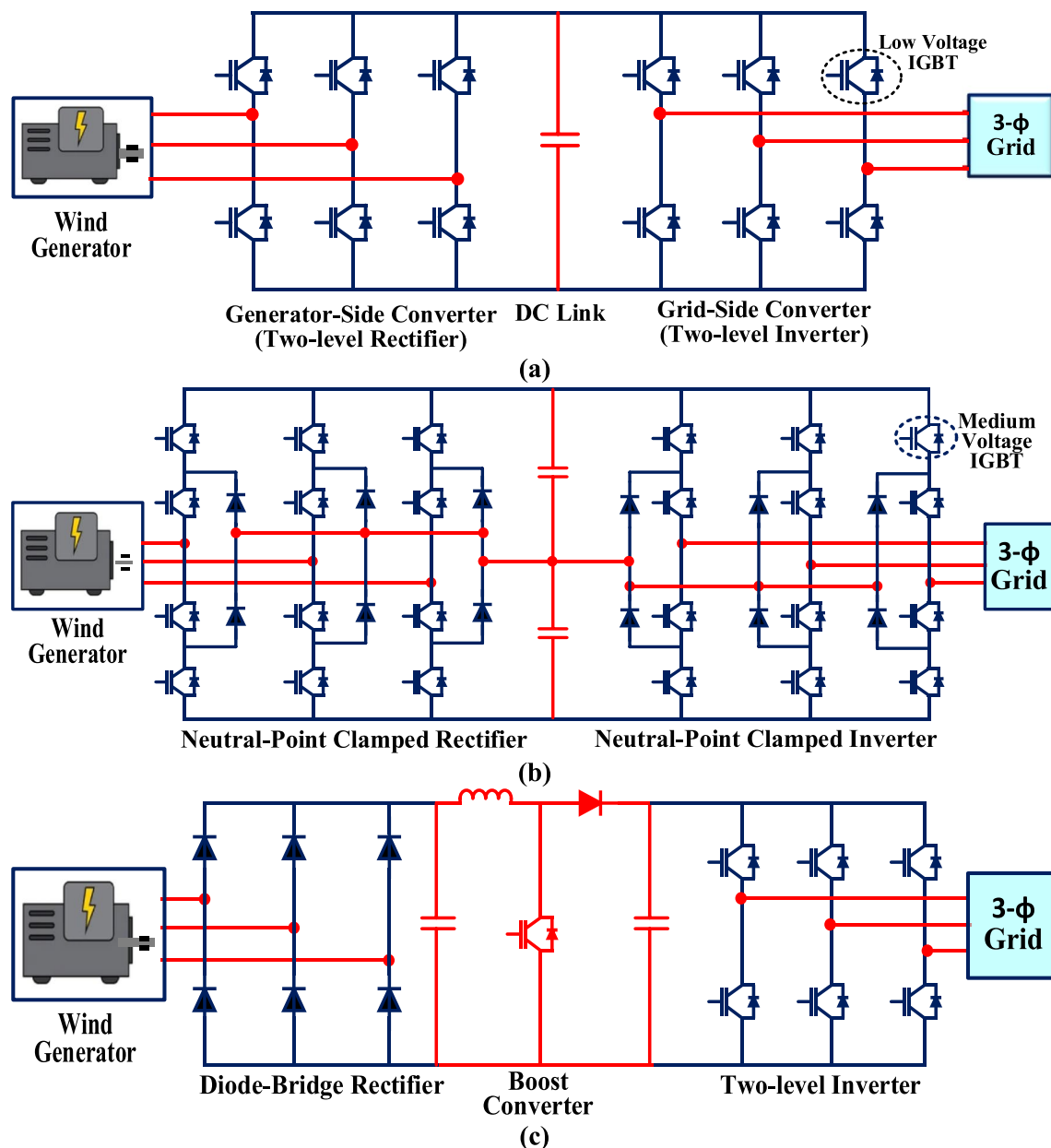


Fig. 5 **a** WECS with two-level back-to-back converter, **b** WECS with three-level back to-back neutral-point clamped converter, **c** WECS with passive generator-side converter

power density which are needed for the wind turbine manufacturers due to the limited space in nacelle. Therefore, development activities and research in the medium voltage power converters area must become more rigorous in the next years [9, 63, 64].

3.2 Passive Generator-Side Power Converters

WECS has a unidirectional power flow from the generator to the grid. Therefore, diode-bridge rectifiers can be used instead of pulse width modulated active converters

to convert AC to DC on the generator side. Diode-bridge rectifiers are less costly and more reliability than pulse width modulated converters. Diode-bridge rectifiers are suitable on the generator side for WECSs which uses EESG and PMSG because of generating the rotor flux by rotor field excitation and permanent magnets, respectively. But they are not suitable for WECSs that use induction generators like SCIG and DFIG because they require magnetizing current during their operation. DC-DC converter is used to enable MPPT operation in WECSs which employing a diode-bridge rectifier. Therefore, there are

three power converters, AC/DC then DC/DC then DC/AC as shown in Fig. 5c [9, 63–65].

3.3 Power Converters Without an Intermediate DC Link

The variable frequency and voltage on the generator side can be transformed to fixed frequency and voltage on the grid side without the need of any DC-link elements which have short life time and are bulky. Matrix converters convert direct AC/AC power through single stage, resulting in high efficiency power conversion with compact size. Additionally, matrix converters provide reliable option for wind turbines, especially when employing in offshore. They can be used with both synchronous and induction wind generators because they enable the four-quadrant operation similar to back-to-back power converters. Direct AC-AC conversion is also performed by cycloconverters where, their output voltage and frequency are only a portion of the input voltage and frequency. They can produce a narrow range of output voltages and frequencies compared with matrix converters [9, 63–65].

4 Grid-Connected Wind Systems

This section will cover an overview of grid-connected wind system configurations. The power electronic converters and electrical generators employed determine the method through which a wind turbine is linked to the grid. The turbine configurations may be classified into a few topologies based on the use of power converters in the WECS: directly connected to the grid without any need for power converters, connection to the grid through a full-scale converter, and connection to the grid through a partial-scale converter. For full-scale converter, the converter rating is equal to generator rating where, whole power exchanged from the generator to grid must be passed via the converter. For partial-scale converter, the converter rating is a proportion of the generator rating where, a part of power exchanged from the generator to grid must be passed via the converter [4, 66]. A broad range of WECS configurations may be produced by combining electrical generator and power converter in various designs and combinations. According to surveys, over the past three decades, there are five common configurations as depicted in Fig. 5. Table 2 summarises the five different types of WECS configurations according to generator used, the utilization of power converters, power converter ratings,

Table 2 The major differences between five types of WECS configuration

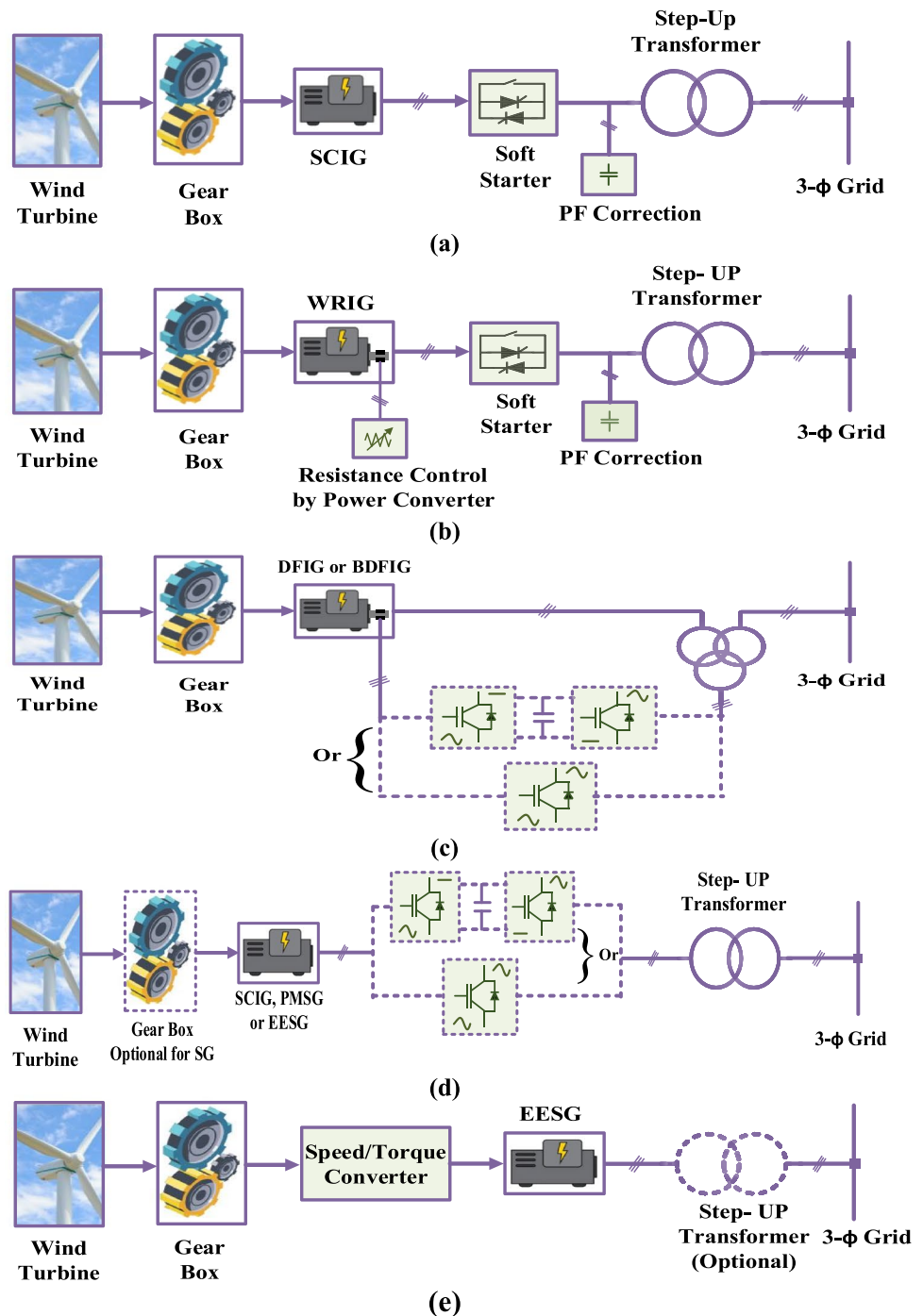
	Type 1	Type 2	Type 3	Type 4	Type 5
Generator used	SCIG	WRIG	DFIG or BDFIG	SCIG, EESG or PMSG	EESG
Power converter utilization	No	Diode and Chopper	(ACtoDCtoAC) or (AC to AC)	(ACtoDCtoAC) or (AC to AC)	No
Power converter ratings	0%	10%	30%	100%	100%
Speed range	$\pm 1\%$ Fixed speed	$\pm 10\%$ Semi variable speed	$\pm 30\%$ Semi variable speed	0–100% Full variable speed	0–100% Full variable speed
Gearbox requirement	Required	Required	Required	Optional for EESG or PMSG	Required
Soft starter	Required	No required	No required	Not required	Not required
PF correction	Required	Not required	Not required	Not required	Not required
MPPT ability	Impossible	Restricted	Achievable	Achievable	Achievable
FRT compliance	By external hardware	By external hardware	By power converter	By power converter	Unknown
Advantages	• High reliability • Low initial cost • Simplicity	• High efficiency • Low maintenance • Long life cycle	• High efficiency • Flexibility against system disturbances • Improved dynamic performance	• Smooth connection to grid • Highest efficiency • More robust against the faults	• Low total system cost and area
Disadvantages	• Direct effect on the grid • Low efficiency	• High energy losses in rotor resistance	• High cost • Required regular maintenance	• High cost • Large size • Complexity • High power converter losses	• Mechanical problems of converter

possible speed range, gearbox requirement, soft-starter need, external reactive power compensation requirement, the ability of MPPT, FRT compliance, advantages and disadvantages. In general, type 3 and type 4 WECS configurations are the best suitable for megawatt scale applications. But, using fully rated power converter for type 4 is the particular disadvantage of this configuration. Therefore, type 3 configuration is considered the best system compared to other systems.

4.1 Type 1 WECS Configuration

Type 1 WECS is the original and earliest wind turbine technology. For this type, SCIG is connected to the grid via a step-up transformer and soft starter without the use of power converter as illustrated in Fig. 6a. The soft starter is bypassed by a switch after the start-up operation and the system continues its operation without the use of any power converter. Three-phase capacitor banks are commonly used

Fig. 6 WECS configurations **a** Type 1, **b** Type 2, **c** Type 3, **d** Type 4, **e** Type 5



to improve PF. This configuration is distinguished by its reliability, low initial costs and simplicity. The main drawbacks of this Type configuration are the direct effect of wind speed fluctuations on the grid, low efficiency of wind energy conversion and the stress on the mechanical wind turbine components during the grid faults. Additionally, this type of WECS has a fixed speed. For compliance with grid standards, fixed-speed WECSs are equipped with extra hardware, like STATCOM. Fixed-speed WECSs were common until a decade ago, and due to its inherent drawbacks, this technology is progressively becoming obsolete [4, 8, 10].

4.2 Type 2 WECS Configuration

Type 2 WECS configuration employs WRIG with variable resistance, which is adjusted by a power converter that is partially rated (10%) as shown in Fig. 6b. This type also necessitates the use of a soft starter, gearbox, and reactive power adjustment. The speed and torque characteristics of generator are affected by the rotor resistance variation, allowing the turbine to run at different speeds. Typically, the speed variation range is restricted to roughly 10% of the rated speed. Therefore, it is a semi-variable-speed WECS configuration. The variable-speed operation decreases mechanical stress, improves the energy conversion efficiency, reduces the requirements of maintenance, and decreases the tear and wear of bearings and gearbox therefore extending the life cycle. But the major problem is high energy losses in the resistance of rotor. Due to its restricted speed range and high losses, this type is becoming less popular among the manufacturers of wind turbines [4, 8, 10].

4.3 Type 3 WECS Configuration

The generator power is supplied to the grid via the rotor and stator windings in type 3 WECS configuration as shown in Fig. 6c. It employs DFIG or BDFIG with a partially rated power converter that is used in the rotor circuit to process the slip power, which is about 30% of the generator rated power. Power converter provides bidirectional power flow in the rotor circuit and extends the generator speed range to about 30%. Therefore, it is a semi-variable-speed WECS configuration. This configuration employs the gearbox in the same way as type 1 and type 2 WECSs, but there is no requirement for a reactive power adjustment or soft starting. In comparison to type 1 and type 2, this system increases the efficiency of power conversion by using MPPT, resilience against power system disruptions and improves dynamic performance. The disadvantages are the high cost due to power converters and the regular maintenance due to gear system [4, 8, 10].

4.4 Type 4 WECS Configuration

The generator power is supplied to the grid via full-scale power converter that is 100% of the rated power of generator for type 4 WECS configuration as shown in Fig. 6d. The employment of full-scale power converters can significantly improve the WECS performance. This type uses SCIG, PMSG or EESG with power rating of up to several megawatts. Because power converter should be rated at the same level as the capacity of generator, the cost, size and complexity of system increase. Although the cost of a power converter is significant, it only accounts for about 7–12% of the total cost of a wind turbine. Furthermore, power converter losses are greater, resulting in poorer efficiency. The generator, on the other hand, is totally disconnected from the grid and can run at full speed range from 0 to 100% with the use of full-scale power converter. In addition, the power converters allow the system to compensate reactive power and to connect with grid smoothly. The merits of this configuration in comparison to other types of WECSs are highest overall wind energy conversion efficiency, more robust against the faults of power system, the necessity for gearbox can be removed by employing synchronous generator with high pole number, and the accomplishment of the best FRT compliance without the use of any additional hardware. Although SCIG and EESG may be utilized in this type, PMSG is the best choice since it removes the requirement for brushes and slip rings, resulting in a more uncomplicated design [4, 8, 10, 67].

4.5 Type 5 WECS Configuration

Type 5 WECS configuration uses EESG which its power is supplied directly to the grid by using torque/speed converter as illustrated in Fig. 6e. The mechanical converter, instead of the electrical converter, is used to provide variable speed operation in wind turbines, which is an old conception. The torque/speed converter, commonly known as variable ratio transmission. It transforms variable wind turbine speed to fixed speed. The generator runs at a constant speed and is linked to the grid directly through synchronizing circuit breaker point without using any step-up transformer or power electronic converter. The total system cost and area are reduced compared to other types because no power electronic converter is required. Despite its benefits, this configuration is rarely employed in WECSs because of the limited knowledge and the problems associated with mechanical converters [10].

5 Grid Codes

Compliance with grid codes is vital because it is necessary to keep the power system running smoothly and safely. The fundamental mains code rules govern the behaviour of

wind turbines in the case of voltage, frequency, and mains failures. The operators of transmission system face several challenges as a result of the integration of large capacity WECS into electrical grids like the reliability and stability of electrical system and power quality. As a result, a number of precise technical criteria known as grid codes which have been updated on a regular basis. The principal keys for grid codes are reactive power control to adjust the voltage of grid, active power control to regulate the frequency of grid, grid power quality, harmonic oscillations, system protection, and the capability of FRT. The comparison between the requirements of group of countries around the world to integrate wind energy into their national grids is shown in Table 3. The major requirements

of grid code are outlined in the following subsections as follows [15, 68]:

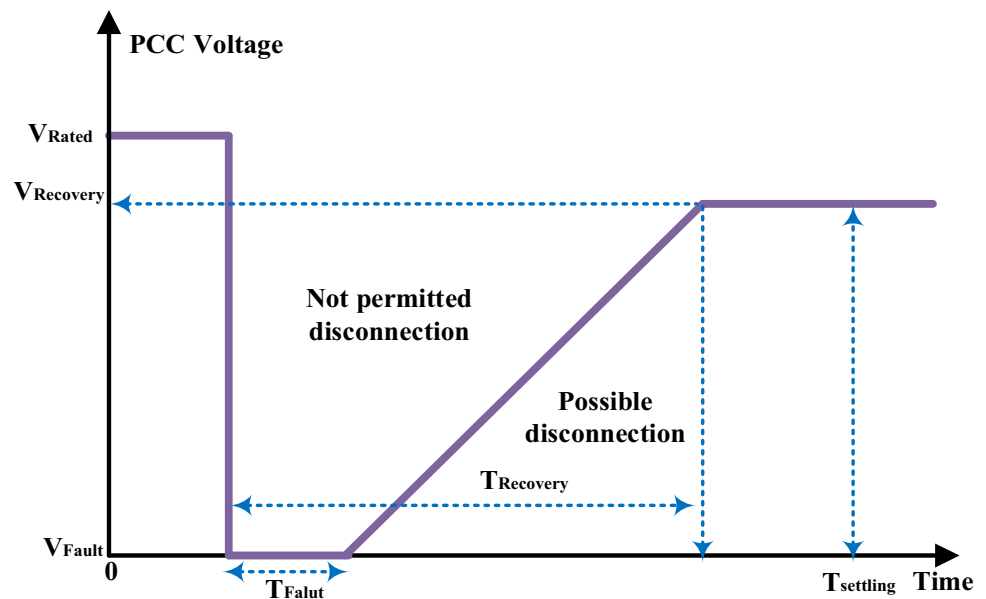
5.1 Frequency Requirements

Most grid codes necessitate that wind turbines operate continuously within a small frequency range, with disconnection permitted only when the frequency falls below the minimum or exceeds the limit of maximum frequency. The power system must meet various frequency range requirements and assess the state of each wind turbine. The control of active power enables the wind turbine to control the generation of real power as demanded by transmission system operators either by frequency control (primary control) or power

Table 3 The comparison between grid codes for group of countries [15, 68]

Country	Frequency requirement		PF requirement		LVRT requirement			
	Frequency limit (HZ)	Maximum duration	Cap. (lead)	Ind. (Lag)	During fault		After fault	
					V _{min} PU	t _{max} Sec	V _{min} PU	t _{max} Sec
Denmark	48.5 < f < 51	Continuous	0.95	0.95	0.2	0.5	0.9	1.5
	48 < f < 51	25 min						
	47.5 < f < 52	5 min						
	47 < f < 52	10 s						
Australia	49.5 < f < 50.5	Continuous	0.93	0.93	0	0.1	0.7	2
	49 < f < 51	10 min						
	48 < f < 51	2 min						
	47.5 < f < 52	9 s						
Canada	59.4 < f < 60.6	Continuous	0.9	0.95	0	0.15	0.85	1
	58.5 < f < 61.5	11 min						
	57.5 < f < 61.7	1.5 min						
	57 < f < 61.7	10 s						
	56.5 < f < 61.7	2 s						
	55.5 < f < 61.7	0.35 s						
Germany	49 < f < 50.5	Continuous	0.95	0.925	0	0.15	0.9	1.5
	48.5 < f < 51.5	30 min						
	47.5 < f < 51.5	10 min						
	46.5 < f < 53.5	10 s						
Ireland	49.5 < f < 50.5	Continuous	0.95	0.95	0.15	0.625	0.9	3
	47.5 < f < 52	60 min						
	47 < f < 52	20 s						
United Kingdom	47.5 < f < 52	Continuous	0.95	0.95	0.15	0.14	0.8	1.21
	47 < f < 52	20 s						
Spain	51.0 < f < 51.5	Continuous	0.91	0.91	0	0.15	0.85	1
	48.0 < f < 51.5	Continuous						
	47.5 < f < 48.0	3 s						
USA	60 < f < 59.5	Continuous	0.95	0.95	0	0.15	0.9	1.75
	59.5 < f < 59.3	10 min						
	59.3 < f < 58.7	10 s						
India	49.5 < f < 50.5	Continuous	No information		0.15	0.3	0.85	3
	47.5 < f < 51.5	WECS remains connected						

Fig. 7 General LVRT capability curve for WECS



curtailment (secondary control). According to the deviations of frequency, the generation of active power is regulated in frequency control. The profile of active power curtailment for various countries varies from each other [9, 68].

5.2 Power Factor Requirements

The requirements of reactive power for wind power plants concern with the ranges of P–Q capability in different control modes. The control of reactive power aids in the compensation of transmission equipments like transformers and cables while also preserving voltage stability. As a result, PF requirement becomes a vital requirement in order to maintain efficient and reliable distribution and transmission networks. Various countries have different approaches for controlling reactive power where, the grid voltage varies depending on the short circuit impedance and power of the circuit. The detailed requirements may differ depending on the country and the capacity of wind power plant. The reactive power control mode, PF control mode, and voltage control mode are the three control modes commonly required by European grid codes. The majority of countries specify the range of reactive power requirement between 0.95 lagging to 0.95 leading, as indicated in Table 3 [9, 68].

5.3 Power Quality and Harmonic Requirements

With respect to power quality requirements, Denmark has the majority of detailed power quality compliance requirements for WECSs. The requirements of power quality for WECSs are not clearly and well defined, thus the same requirements that are applied on the traditional power grids are used. For example, USA follows IEEE 519 while, Italy

and Ireland apply IEC standards and China uses GB/T 14549 requirements. The Danish grid code provides higher standards for power quality requirements for WECSs with extra limits for total and partial weighted harmonic distortion and short circuit ratio as compared to conventional power plants. The power quality requirements differ according to the capacity of WECSs. For smaller WECSs, harmonic distortions limits are defined. However, harmonic distortion limits for larger WECSs are determined by voltage distortion to accommodate for local changes in grid impedance and WECSs capacity relative to grid capacity [9].

5.4 Fault Ride Through Requirements

The unexpected disconnection of large-scale units of wind power generation during faults causes the utility network to become unstable. Therefore, grid code requirements for FRT capabilities are provided to assure continuous operation, minimize excessive power losses caused by faults, assist grid recovery during a fault and reduce resynchronization issues. The zero-voltage ride through, low voltage ride through (LVRT), and high voltage ride through requirements are all covered by the FRT standard. The requirements of LVRT and zero-voltage ride are nearly identical where, during grid failures, the grid voltage drops to zero in the zero-voltage ride profile, whereas the grid voltage drops from 15 to 25% of its rated value in the LVRT profile. LVRT requirement is described using voltage–time characteristic curves, which also represent the minimum allowable voltage sag. Figure 7 shows a general LVRT capability curve for WECS. The grid codes for LVRT determine the allowable fault voltage (V_{Fault}), fault time (T_{fault}), recovery time ($T_{recovery}$), recovered voltage within recovery time ($V_{Recovery}$) and the settling time

(T_{settling}) as shown in Fig. 6. These LVRT requirements limits may differ from one country to another [68, 69].

The classification of different types of LVRT techniques for grid-connected based WECS is shown in Fig. 8. These techniques may be classified into two groups. The first sort contains hardware modifications, while the second type includes changes to the traditional controllers for DFIG. There are two types of hardware-based LVRT methods, the first method employs protection circuits, while the second employs reactive power injection devices [68, 69].

The crowbar protection is the most commonly known hardware-based LVRT technique for DFIG based WECS. It is an electrical circuit made up of resistors that are linked across the terminals of rotor winding. Figure 9a depicts the configuration of LVRT capability using crowbar protection for DFIG based WECS. The enhancement of crowbar protection circuit is performed by combining the series RL circuit and crowbar resistance to overcome the negative effects

of crowbar resistance circuit or by integrating the crowbar circuit with dynamic series resistance [68–70]. Passive resistive hardware known as a stator damping resistor (SDR) is connected to the winding of stator for DFIG in a cascaded connection. As shown in Fig. 9b, three resistors are connected in parallel with a static bidirectional bypass switch. The key advantage of SDR is that, unlike the crowbar protection strategy, the generator will not be disconnected and will continue to deliver power during disturbances that improve grid stability [68, 70]. The DC-link voltage may increase over the maximum allowable limits during fault occurrences therefore, the power electronic switches and DC-link capacitor may damage. This problem may be solved by cascading chopper circuit to the DC-link capacitor as shown in Fig. 9c. The principal disadvantage of this method is that it wastes energy rather than storing it such as an energy storage system, although it is less expensive than a storage system [68, 71]. Dynamic voltage restorer (DVR) comprises a

Fig. 8 LVRT approaches classification for grid-connected based WECS

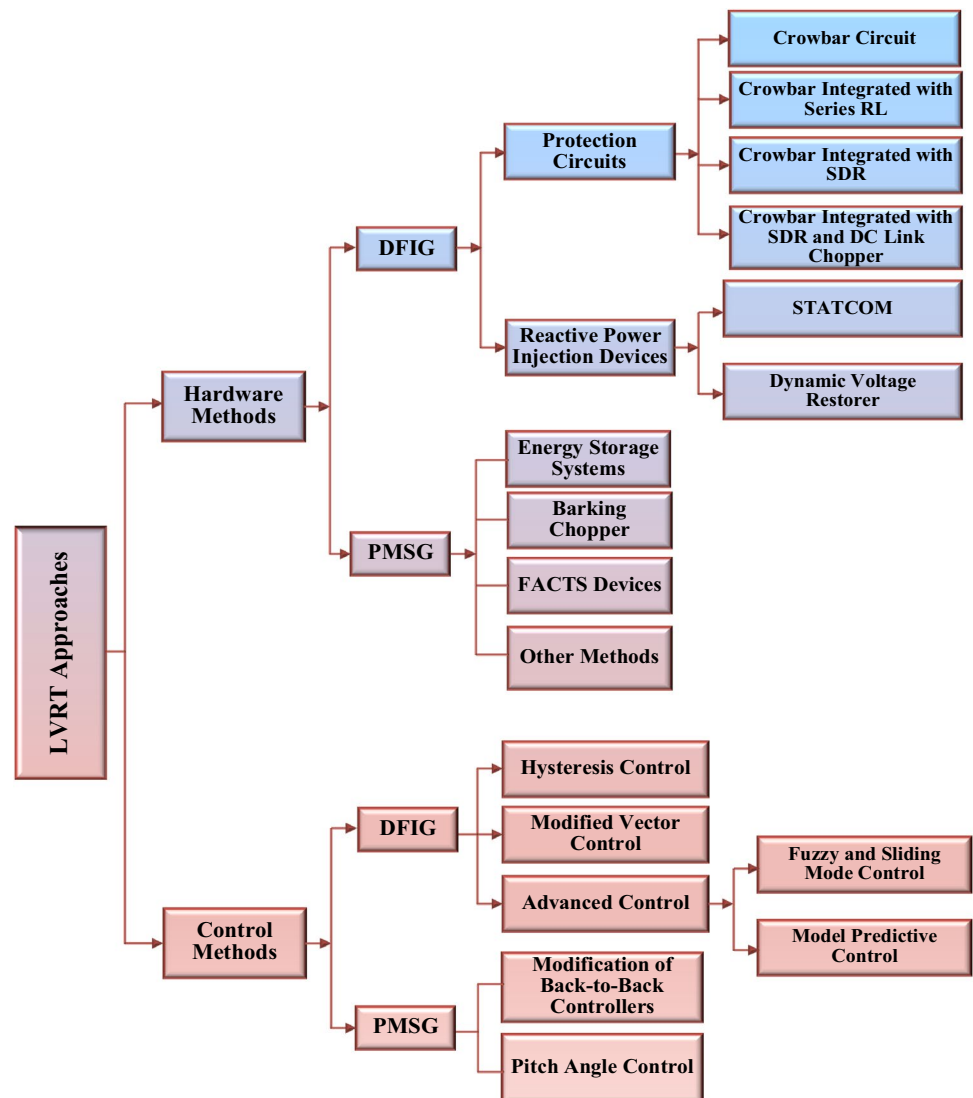
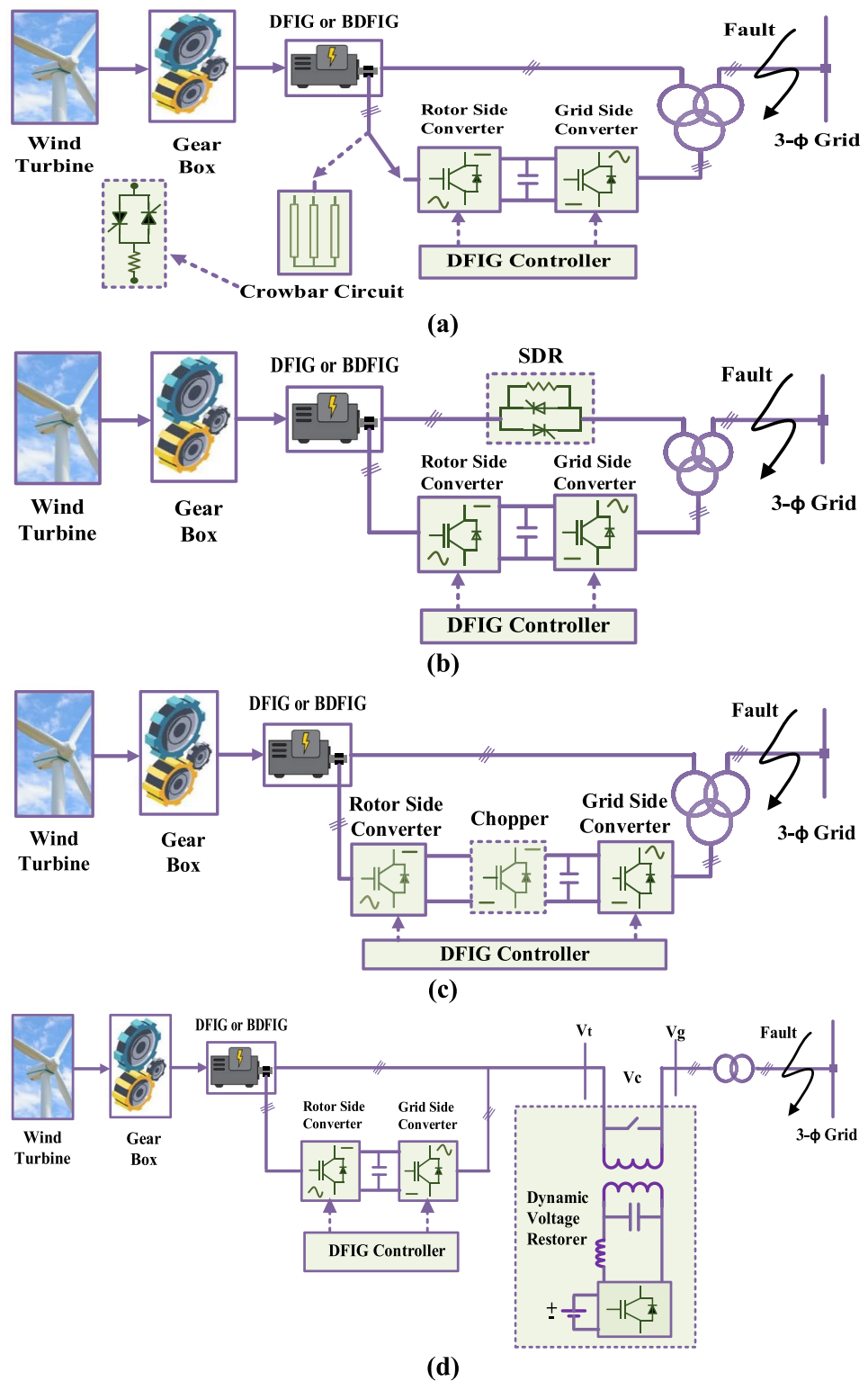


Fig. 9 LVRT capability for DFIG based WECS using **a** crowbar protection, **b** SDR, **c** DC-link chopper, **d** DVR



voltage-source converter that is placed in series between grid and DFIG to enhance LVRT requirement. DVR can compensate the sag and swell of voltage and harmonics. DVR is connected in series at point of common coupling (PCC) for voltage injection, allowing for the adjustment of reactive

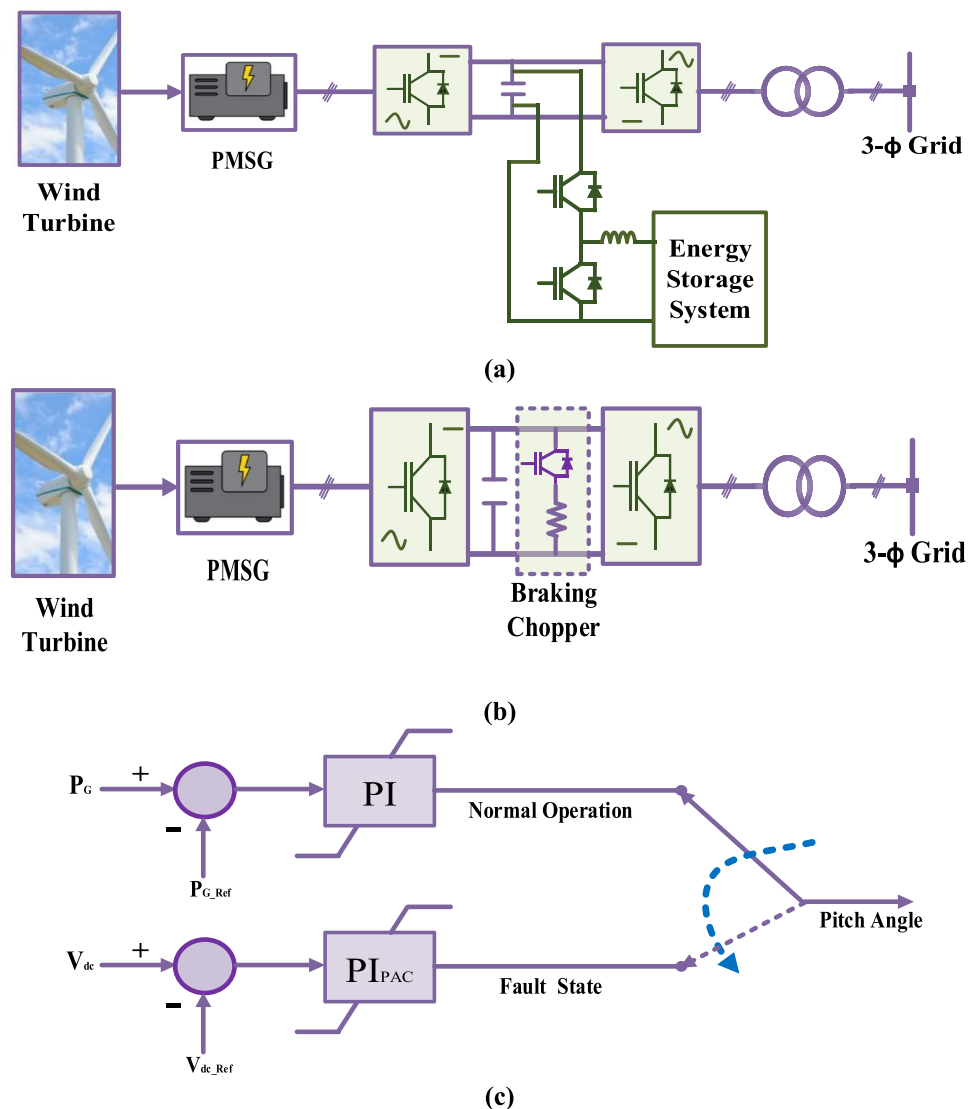
and active power in the grid, that aids in fault recovery as illustrated in Fig. 9d [68, 72].

For PMSG based WECS, power fed into the grid begins to decrease during grid failures. As a result, the grid-side converter controller is unable to detect voltage drop at PCC,

and the generator-side converter stays to supply real power to the DC-link capacitor. This active power imbalance raises DC-link voltage, which may damage capacitors causing the saturation of generator, and increasing the stress of voltage on the grid-side and generator-side converters. To overcome these problems, LVRT methods are used [68, 69, 73]. As shown in Fig. 10a, the energy storage system (ESS) is connected to DC link through a buck-boost converter for LVRT improvement. ESS absorbs the excess energy of the DC link during fault, preventing the DC link from overvoltage. The most popular types of ESSs used for enhancement of LVRT capability are batteries, flywheels, electrical double-layer capacitors, and superconducting magnetic ESS [68, 74]. LVRT capability using braking chopper is an active crowbar circuit that relies on external devices for LVRT enhancement. It comprises a switch and a high-power resistance connected in parallel with DC link of PMSG as shown in Fig. 10b. It features cheap cost and a simple control

structure [68, 73]. The pitch angle control (PAC) is recently implemented to enhance the LVRT for PMSG based WECS through some modifications to the controller. PAC is generally employed to reduce the input of power in the event of gusty wind. Figure 10c shows a simple PAC for the capability of LVRT for PMSG based WECS. The normal control mode will stay active for all wind speeds greater than the rated value. The PI controller, which operates with regard to voltage changes, is employed to adjust the value of pitch angle reference during instances of drop in grid voltage. The major drawback of this scheme is slow response to the dips of grid-side voltage but, the main advantage of this approach is quite inexpensive [68]. Flexible AC transmission system (FACTS) is an advanced technology that depends on the power electronic devices. FACTS has been utilized to protect sensitive loads with great effectiveness from damping and transient oscillations, as well as the sags of voltage. Lately, FACTS devices have developed as a viable option

Fig. 10 LVRT capability for PMSG based WECS using **a** ESS, **b** Braking chopper, **c** PAC



for keeping wind turbine systems linked to the electrical grid in the event of fault occurrences. These devices are classified into three types with regard to their connection, which includes shunt, series, and hybrid connections. The common types of FACTS devices used in PMSG based WECS for LVRT requirements are DVR, static VAR compensator (SVC), static synchronous compensator (STATCOM), and unified power flow controller (UPFC). The arrangement of LVRT capability using different types of FACTS devices for PMSG based WECS is illustrated in Fig. 11. For PMSG LVRT applications, the most efficient FACTS device is the UPFC, despite its high cost [68, 73].

5.4.1 Recent LVRT Methods

In recent years, cost-effective LVRT approaches for WECSs have been introduced. Table 4 presents recent modified

control methods for enhancing LVRT capability. This subsection presents some interesting background papers about improving LVRT during the past 4 years as follows:

In [75], Priyavarthini et al. presented a novel approach for usage of DVR to adjust reactive power requirement for the fixed speed wind generator along with FRT regardless of reducing the swell and sag of grid voltage. The mitigation of swell and sag was performed by proposed phase angle control. The key features of the proposed control are maintaining constant phase shift between the voltage of grid and generator, better DVR usage is obtained without affecting on wind generator steady state operation, and no effect on the active power provided by generator to the grid. In [76], Yan et al. introduced a non-linear adaptive controller for the ESS embedded DVR in order to enhance LVRT requirements for WECSs. The ESS embedded DVR was controlled by proposed control to rapidly compensate the dips of grid voltage for supporting the voltage

Fig. 11 LVRT capability for PMSG based WECS using FACTS devices: **a** STATCOM, **b** SVR, **c** DVR, **d** UPFC

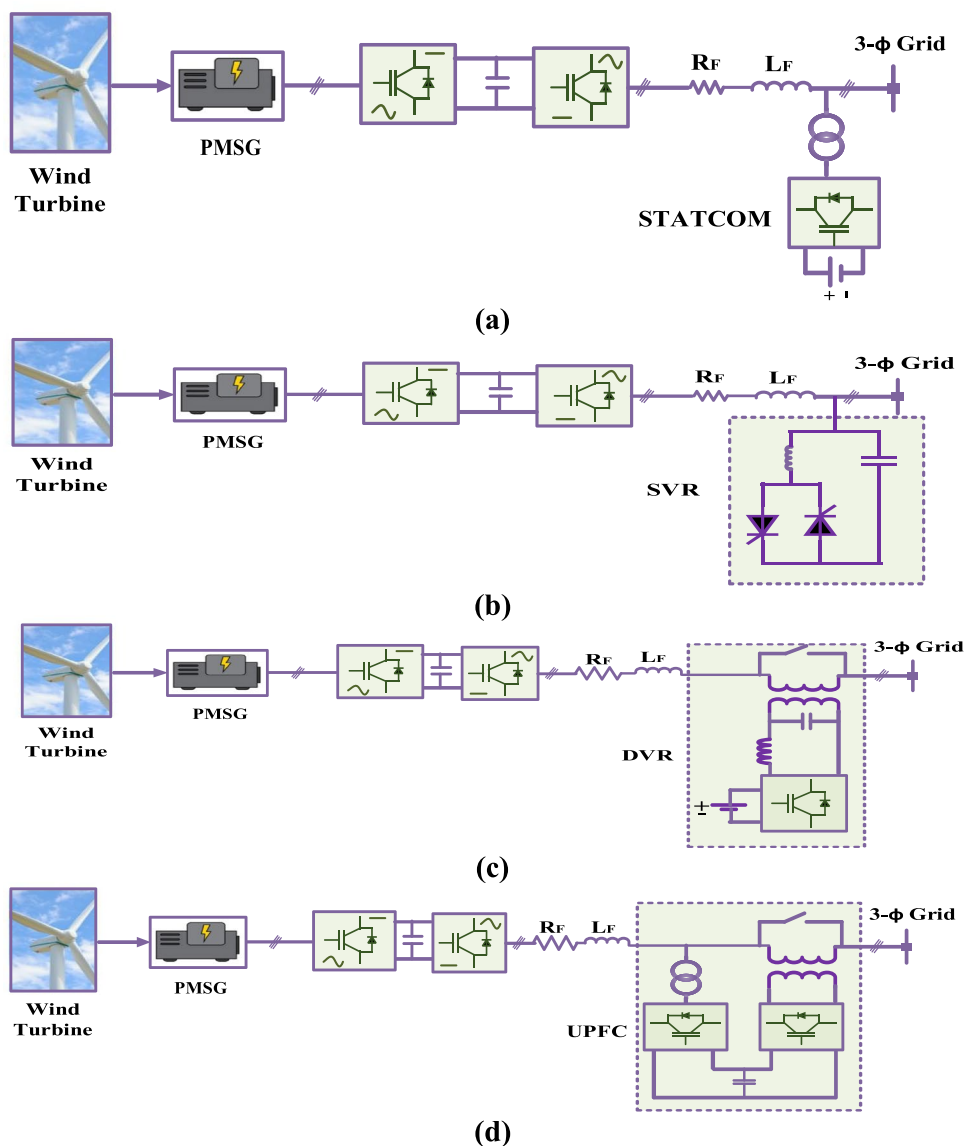


Table 4 Recent modified control methods for enhancing LVRT capability

Refs.	Year	Improvement method	LVRT	Advantages
[75]	2018	Implementing phase angle control	DVR	• Provide a steady operation of constant reactive and real power from and to the wind generator to grid even under unsymmetrical grid voltage condition
[76]	2018	Non-linear adaptive control	ESS embedded DVR	• Independent of full-state feedbacks and precise system model
[77]	2019	Nonlinear control theory arguments	Modified of back-to-back controllers	• Providing robustness and additional oscillations damping • With minimal assistance of additional protective hardware
[78]	2019	Coordinated control	Modified of back-to-back controllers	• Simplicity • No oscillation in the active power • Very fast in removing faults
[79]	2020	Modified feedback control	DVR	• Stable and very effective • Highly reliable
[80]	2020	Model-free adaptive control	Modified unified power flow controller	• Rapid response, robustness, and simple implementation under several disturbances and operating conditions
[81]	2020	Modified switch type fault current limiter	Switch type fault current limiter	• Cost effective • Requires little maintenance
[82]	2020	Modified rotor-side converter control	Modified control method	• Simple implementation • Increase system stability and reliability • Cost-effective
[83]	2021	PI controller for off-delay angle control	Modified control method	• Keeping the PCC voltage in allowable levels during faults
[84]	2021	Reduced order generalized integrator based negative-positive sequence control	Modified control method	• Enhancing active and reactive power, PCC voltage, rotor and stator currents, and DC-link voltage under balanced and under unbalanced fault conditions
[85]	2021	Modified demagnetization control	Modified control method	• Better dynamic responses and improving the security operations and power system stability • Comply LVRT capability with the higher penetration of wind generation
[86]	2021	Braking chopper control approach based on sliding mode control	Braking chopper	• Robustness and effectuality • Cost-effective and reliable LVRT improvement method
[87]	2021	Hybrid algorithm based on random forest and modified elephant herding techniques	Control method	• Enhancing power quality issues
[88]	2021	Novel modified bridge resistive type	Superconducting fault current limiter	• Reducing the steady-state operating loss • Controlling the value of resistance

of PCC and store the blocked power of wind to overcome the possible variations of wind energy. In [77], Tilli et al. proposed a novel control technique for the rotor-side converter for DFIGs to endow WECS with efficient LVRT capability under severe unbalanced and balanced grid voltage decreases. The proposed control was depended on the arguments of non-linear control theory which were employed for the design of controller to mitigate oscillations of rotor voltages and currents where, the system will not disconnect for protection. In [78], Tripathi et al. presented a simple control approach for the improving of LVRT capability for PMSG based WECS. The major idea of this approach was that the altering between the

variables of reference control of the grid-side and stator-side converters for keeping the DC-link voltage in the safe range during the occurrence of grid voltage drops by storing excess real power in the inertia of system. In [79], Hossam-Eldin et al. introduced the parallel switched DVR with modified feedback control for the improvement of FRT capability in DFIG based WECS under abnormal conditions. In [80], Mosaad et al. presented a model-free adaptive control for UPFC for improving the FRT capability of the DFIG based WECS during several disturbance events and the overall dynamic performance during wind gusts. The proposed control is employed to control the shunt and series converters of the unified power flow

controller. In [81], Asghar et al. presented a novel structure of modified switch type fault current limiter for enhancing LVRT capability of the DFIG based WECS. The proposed system involves of two parts. The first part is diode bridge, IGBT switch and snubber circuit while the second part is the major part, which consists of a limiting inductor connected in series with the main line. In [82], Ali et al. introduced a modified control for rotor-side converter to improve LVRT capability for DFIG based WECSs during the occurrence of grid faults. This research focused on the reduction of DC-link voltage fluctuations and rotor overcurrents. To improve the transient performance for the rotor-side converter control, additional voltage terms were injected into the references of rotor voltage. In [83], El-Naggar et al. proposed an optimum PI controller for 4-phase 8/6 switched reluctance generator to control the off-delay angle to improve LVRT capability for WECS during the variations of wind speed and faults. In [84], Raghavendran et al. presented reduced order generalized integrator based negative–positive sequence control with the purpose of controlling grid-side converter of PMSG for FRT capability in order to preserve the parameters of system stable under faults. In [85], Senapati et al. introduced an adapted demagnetization control scheme and an external resistor on the rotor and stator side of DFIG to enhance the LVRT capability. By lowering the time constant, the external resistance in the stator accelerates the transient flux damping. In [86], Nasiri et al. introduced a control approach based on sliding mode control for braking chopper during grid faults and normal conditions for PMSG based WECS. In [87], Manohar et al. introduced a hybrid technique control model for improving the LVRT capability of DFIG. The hybrid technique was the combined implementation of both random forest and modified elephant herding algorithms.

The objective function that considers multiple parameters related to LVRT such as active and reactive power, voltages, and currents was solved by modified elephant herding method. While, the random forest technique was used to predict the optimal control signals of generator-side converter. In [88], Du et al. proposed a novel modified bridge-resistive-type superconducting fault current limiter for the enhancement of FRT capability for DFIG. The comparative analysis between the traditional superconducting fault current limiter and the proposed limiter shows that the proposed method can enhance LVRT capability during the fault occurrence and after its removal.

6 Electrical Controllers

This section presents an overall view of electrical controllers that are used for WECSs, such as converter controllers, MPPT controller and frequency controller.

6.1 Converter Controllers

The controllers for generator-side and grid-side converters are the major controllers that are used in grid connected WECSs. These conventional controllers are discussed as follows:

6.1.1 Generator-Side Converter Control

The purposes of the controller for generator-side converter are to perform perfect decoupling control of the machine flux and torque in order to regulate its speed, extract maximum power from wind, and regulate the active and reactive components of machine current. Direct torque control and field-oriented control are two major control types used to achieve these goals in an efficient way [15, 89]. Blachke introduced field-oriented control in 1969 to regulate the generator speed by changing the amplitude, phase, and frequency of stator current. This approach has a number of advantages, including low reactive and active power ripple, good steady state performance, minimal switching loss, constant and low converter switching frequency, and robustness against measurement noise. However, it has a number of drawbacks, including a lot of online calculation because of pulse width modulation, slow transient response under grid failure circumstances because of limited control bandwidth of PI controllers, dependence on the grid filter parameters change, the difficulties of parameter tuning, and reducing the robustness under varying operating conditions [15, 89, 90]. Direct torque control has many advantages as compared with vector control like no requirement for coordinating transformations that lead to decreasing the compensation burden, quicker dynamic response, no requirement for an internal current control loop, perfectly decoupled system, and simplicity. The torque of the generator can be varied either through adjusting the torque angle or flux. The major disadvantages of this control are variable switching frequency operation, high current total harmonic distortion, high error of tracking, and large filter inductance. The use of space vector modulation based direct torque control can solve the difficulties of variable switching frequency operation while also reducing the filter inductance size [15, 89, 91].

6.1.2 Grid-Side Converter Control

The functions of grid-side converter are controlling of reactive and real power in the grid, adjusting DC-link voltage, grid synchronization, and ensuring high-quality injected power. Grid-side converter configuration is independent of the type of generator. The voltage-oriented control and direct power control are two types of grid-side converter control [15, 89]. To perform grid synchronization, voltage-oriented control used a synchronous reference frame phase-locked

loop. It has several advantages such as quicker tracking ability, high sensitivity under sudden parametric change, higher injected power quality, no need for voltage vector switching states, and operating in both variable and fixed switching frequency. The drawbacks of employing this type of control are weak dynamic response, complex implementation, and the requirement for co-ordinate transformation block [15, 89]. Direct power control technique is similar to the direct torque control used in the generator-side converter. It uses a voltage vector state for the control of reactive and active power. It has many advantages such as simplicity due to no necessity for an internal current control loop, robustness, rapid transient response, close unity power factor, no need for pulse width modulation and co-ordinate transformation, and stable operation under parametric change. The disadvantages of this method are increasing of total harmonic distortion, high current ripples, and increasing of filter size, and poor performance when the operating point of the converters reaches its rated power [15, 89, 91].

6.1.3 Recent Converter Controllers

Table 5 presents recent control methods for grid-side and generator-side converters during the past four years which are summarized as follows:

In [92], Giannakis et al. presented a sensorless decoupled P-Q control for DFIG based WECS with intelligent controllers. The estimation of the rotor speed and position was achieved by modified phase-locked loop instead of using mechanical sensors to measure them. The whole implemented control method was based on fuzzy logic controllers that were employed for tuning the gains of PI controllers instead of constant gain. In [93], Debouza et al. introduced a disturbance observer based control scheme for grid connected DFIG. The proposed control comprises disturbance observer and state-feedback controller which was used to eliminate the steady state error by compensating for model uncertainty. In [94], Xiong et al. proposed a fractional order sliding mode control based direct torque control approach for grid connected DFIG with the purpose of adjusting the output reactive and real power for DFIG directly. In [95], Harrabi et al. presented a Takagi–Sugeno fuzzy control technique for WECSs. The WECS system was divided into two interrelated subsystems to make modelling easier. After that, each subsystem was represented by a Takagi–Sugeno fuzzy model. In [96], Merabet et al. introduced an overall sliding mode control method for controlling the currents and powers of stator and grid and adjusting DC-link voltage for DFIG based WECS. The improvements of robustness to uncertainties and tracking performance were achieved by sliding mode control law structure, integral and sliding terms, and compensating combination. In [90], Mohammadi et al. proposed combined control scheme based on virtual

flux direct power control and vector control for grid-side converter of DFIG based WECSs to take the advantages of two controllers. In [97], Soliman et al. presented a novel cascaded control approach based on genetic algorithm and adaptive neuro-fuzzy inference system for controlling both the grid-side inverter and the generator-side converter. The proposed control was designed by using the data of dynamic response for the studied system with PI controllers under various wind speed conditions. In [98], Morshed et al. proposed an integral terminal sliding mode-based control for both the rotor-side and grid-side converters of DFIG based WECS under disturbances and unbalanced voltage conditions. In [99], Mbukani and Guleet al. presented a phase-locked loop based high-order sliding mode observer estimator for the sensorless vector control of the rotor of DFIG based WECS using a slip speed and angle estimator. In [100], Djilali et al. introduced a real-time neural sliding mode field oriented control for DFIG based WECS connected to the grid. This control approach relies on high-order recurrent neural network identifier which is trained online by an extended Kalman filter which is employed to approximate DFIG mathematical models and DC link. In [101], Abo-Khalil et al. presented a multivariable state feedback current controller with a feedforward component for enhancing the transient and steady-state characteristics of controller. Multivariable state feedback controllers were proposed to substitute the PI controllers on the grid-side and rotor-side converters. In [102], Moradi et al. proposed an adaptive sliding mode type-2 neuro-fuzzy controller for DFIG power control. Due to the changes in parameters in WECS and uncertainties in wind speed, the proposed control used an interval type-2 fuzzy system. Sliding mode control was employed to train online the membership functions parameters of type-2 fuzzy controller to improve its performance. In [103], Rocha-Osorio et al. introduced a deadbeat-fuzzy controller for improvement of the steady-state response of DFIG based WECS. The proposed control was compared with conventional deadbeat power controller and a classical DFIG PI controller. It was concluded that the proposed deadbeat fuzzy controller is an appropriate solution to the rotor-side control. In [104], Amin and Uddin presented a novel adaptive backstepping based nonlinear control method for the controller of rotor-side converter to reduce machine losses and to overcome parameter uncertainties to improve transient performance and ensure stability of the grid connected DFIG based WECS. In [105], Xiong et al. introduced a fractional order sliding mode control scheme for PMSG to improve the quality of output power. The proposed control was used to control both grid-side and generator-side converters. The boundaries of controller coefficient were decided by stability analysis based on Lyapunov function and then the optimal parameters were obtained by the gravitational search technique. In [106], Samanes et al. proposed

Table 5 Recent control methods for grid-side and generator-side converters

Refs.	Controller type	Year	Goals	Advantages
[92]	Sensorless with fuzzy logic and PI controllers	2018	• The control of real and reactive power, rotor speed for MPPT, and DC-link voltage	• The estimation of the rotor speed and position without using differentiation • Increase the robustness due to the independence against machine parameters variations
[93]	State feedback controller and disturbance observer	2018	• Regulating the stator currents for achieving the direct control of stator real and reactive power	• Good transient performance • Guaranteeing zero steady state error
[94]	Fractional order sliding mode control	2018	• The control of real and reactive power for DFIG based WECS	• High robust performance • Fast convergent velocity • Simplicity to implement
[95]	Takagi–Sugeno fuzzy control	2018	• Extracting the maximum amount of available power • Control of DC bus voltage	• Suitable for complex non-linear systems
[96]	Sliding mode control	2018	• The control of power and current for the stator and grid • DC-link voltage control	• Effective control strategy for non-linear systems subject to uncertainties, parametric mismatch and external unknown disturbances
[90]	Combined vector control and virtual flux direct power control	2018	• Maintaining constant DC link voltage • Adjust the reactive power of grid side	• Faster dynamic responses • Robustness to grid filter parameter variation and to measurement noise • Lower computation • Simple implementation • Less current harmonic distortion • Lower power ripple
[97]	Hybrid ANFIS-GA control	2018	• Control of grid-side inverter and generator-side converter	• Fast convergence and accurate responses • Proper design, high capability, and robustness to deal with non-linear dynamic systems
[98]	Integral terminal sliding mode control	2019	• Enhance the power quality	• Robustness and high-quality transient characteristics • Fast response
[99]	Phase locked loop based high-order sliding mode observer estimator	2019	• Improve the estimation performance	• Starting the estimation of the rotor speed before connecting to a load or to grid
[100]	Real-time neural sliding mode field-oriented control	2019	• DC-link voltage control • Maintain the electric PF at nominal values • Stator real and reactive power control	• Extract the maximum wind power under different fault scenarios • Increase system reliability • Improve quality and quantity of power generation
[101]	Multivariable state feedback current controller	2019	• Improving the dynamic and steady-state characteristics	• The effective cancellation of the disturbances in the rotor current control robustly
[102]	Adaptive sliding mode type-2 neuro fuzzy controller	2019	• Control the active and reactive power	• Better performance for tracking the peak power • More robustness against generator parameter changes
[103]	Deadbeat-fuzzy controller	2019	• Reduce the steady-state error • Enhance the robustness against generator parameter changes	• Better tracking of the reference signals

Table 5 (continued)

Refs.	Controller type	Year	Goals	Advantages
[104]	Adaptive backstepping based nonlinear control	2020	• Stabilizing both the rotor-side and grid-side current control loops • Enhance efficiency by reduction of DFIG losses	• Enhance performance by reducing current and speed errors • Improving DFIG efficiency by decreasing total power losses • Tracking reference rotor speed to extract maximum wind power
[105]	Fractional order sliding mode control	2020	• The enhancement of power quality for PMSG • Regulate PMSG output real and reactive power	• Simple structure and chattering free • Precise tracking performance • Good robust performance • Reduce the harmonics of the system
[106]	Sub-synchronous resonance control	2020	• Sub-synchronous resonances damping • Stabilizing wind farm for any line impedance compensation level	• Robustness • No need for information about the grid
[107]	Fixed-time fractional order sliding mode controller	2021	• Enhance the power quality • Improving the dynamic performances	• Easy to design and implement in practice • Robust and effective for the PMSG wind turbine
[108]	Command-filtered integral backstepping controller	2021	• Rotor-side converter control under various fault conditions	• Less complex • Robustness with different wind speeds
[109]	Gaussian process regression based fuzzy fractional-order terminal sliding mode control	2021	• Solve the problem of inherent chattering • Maintaining the system robustness and stability	• Minimum chattering • Offering fast dynamic response • Higher accuracy • Disturbance rejection
[110]	Adaptive multi-objective sliding mode control	2021	• Fulfill the control goals of both rotor-side converter and grid-side converter	• Robustness • Capable of dealing with extra difficulties such as unmodeled quantities and parameters variations
[111]	Nonlinear adaptive backstepping control	2021	• Active and reactive power control	• Exhibit good performance under a real wind profile • Robustness against variations in the wind profile and generator parameters

a simple sub-synchronous resonance control approach which employed in the rotor-side converter with variable gain in PI controller regulated as a function of DFIG rotational speed. In [107], Huang et al. presented a fixed-time fractional-order sliding mode controller for PMSG based WECS for power quality enhancement. The proposed control was implemented in the rotor-side converter and grid-side converter for stabilizing PMSG based WECS within an upper limited convergence time and for improving the dynamic performance. In [108], Patel et al. introduced a command-filtered integral backstepping controller for controlling the rotor-side converter in DFIG based WECS. By adjusting the electromagnetic torque, this controller basically stabilizes the speed of rotor. The Lyapunov criteria were employed to establish the overall stability of closed loop system and controller parameters were optimally determined by using particle swarm optimization. In [109], Sami et al. presented a Gaussian process regression based fuzzy fractional-order terminal sliding mode control for DFIG based WECS. It is a hybrid of fuzzy control and fractional order control, which solves the issues of inherent chattering in both conventional controllers while preserving system stability and robustness. This research also proposed a new wind speed estimation approach based on Gaussian process regression where, its inputs are rotational speed and wind turbine power. In [110], Noussi et al. introduced an adaptive multi-objective sliding mode control for DFIG based WECS. The proposed control was designed to control both the rotor-side converter and the grid-side converter. The difficulties of the control problem that are represented by external disturbances and the uncertainty and nonlinearity of the system dynamics were solved by the proposed control. In [111], Bossoufi et al. proposed a nonlinear adaptive backstepping control scheme for DFIG based WECS. To ensure the system stability, the backstepping control approach used the Lyapunov nonlinear algorithm. It was employed on both the rotor-side converter and the grid-side converter and then enhanced by using estimators to make the proposed system more robust against parametric change.

6.2 MPPT Controllers

With the growing usage of wind generators in power grids, it is necessary to extract the available maximum wind power. Therefore, WECSs must track or operate at maximum power point (MPP). The wind turbine produces rated power (P_{rated}) at a certain speed of wind (V_{rated}). The control of output power for wind turbine can be performed more efficiently within a certain bounded wind speed range between cut-in speed ($V_{\text{Cut-In}}$) and cut-out speed ($V_{\text{Cut-Out}}$). For safe operation for both the generator and turbine, turbine must not be operated outside of this range. Figure 12a illustrates the four major operational

zones of wind turbine. In the second region, the controller of wind turbine begins to operate to extract the maximum possible wind energy [112, 113]. For a wind turbine exposed to a wind of velocity V_w (m/s), has a blade radius R and a swept area by blades A (m^2); the power extracted from wind can be calculated from:

$$P_m = \frac{1}{2} C_p \rho A V_w^3, \quad (1)$$

where P_m is the output power of turbine (watt), ρ is the air density (kg/m^3), C_p is the power coefficient of turbine which is a function of tip speed ratio (λ). The tip speed ratio of wind turbine is defined as follows where ω_m is the mechanical rotational speed of the turbine (rad/s):

$$\lambda = \frac{\omega_m R}{V_w} \quad (2)$$

From the equations from (1) to (2), the output power is primarily dependent on turbine power coefficient value and this depends on wind turbine rotor speed. For a certain wind speed, the optimal power coefficient values and blade tip speed ratio occur at only one optimal rotor speed for maximizing the extracted power. For operating the system at MPP in all operational circumstances, the turbine rotational speed should be changed as the speed of wind varies instantly.

The nonlinear power-speed characteristics of a turbine at various wind speeds is depicted in Fig. 12b. The wind turbine output power can be maximized by regulating the rotational speed of turbine rotor at optimal value, and maintaining operating points at optimal power coefficient value [112, 113]. Figure 13 shows the classification of MPPT techniques which are divided into main categories in general; indirect power controller, direct power controller, hybrid MPPT techniques, and smart MPPT techniques. The indirect power controller based MPPT algorithms increase the output power by adjusting the mechanical power according to pre-stored data of power-speed curves. While, the direct power controller based MPPT algorithms analyze the alteration of electrical output power for tracking the MPP. In this subsection, a brief review of several MPPT algorithms is discussed and is summarized in Table 6 [112–114].

6.2.1 Indirect Power Controller Based MPPT Algorithms

6.2.1.1 Power Signal Feedback Algorithm This method necessitates knowledge about the curve of maximum power for a wind turbine as shown in Fig. 12b, which is followed by its control methods. Experimental tests or simulations can be used to obtain maximum power curves for individual wind turbines. The block diagram of power signal feedback (PSF) MPPT algorithm for WECS is shown in Fig. 14a. Look-up table is employed in this algorithm for record-

Fig. 12 **a** The operating regions of wind turbine, **b** Turbine characteristic for various wind speeds

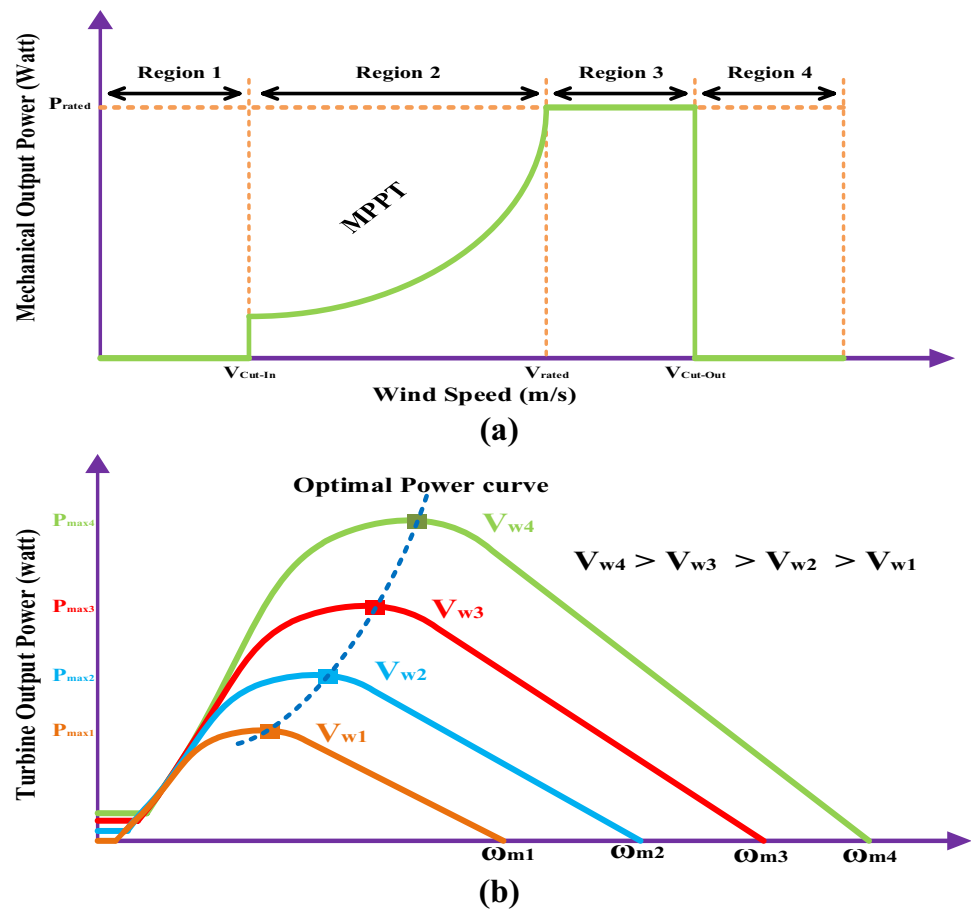


Fig. 13 The classification of MPPT approaches

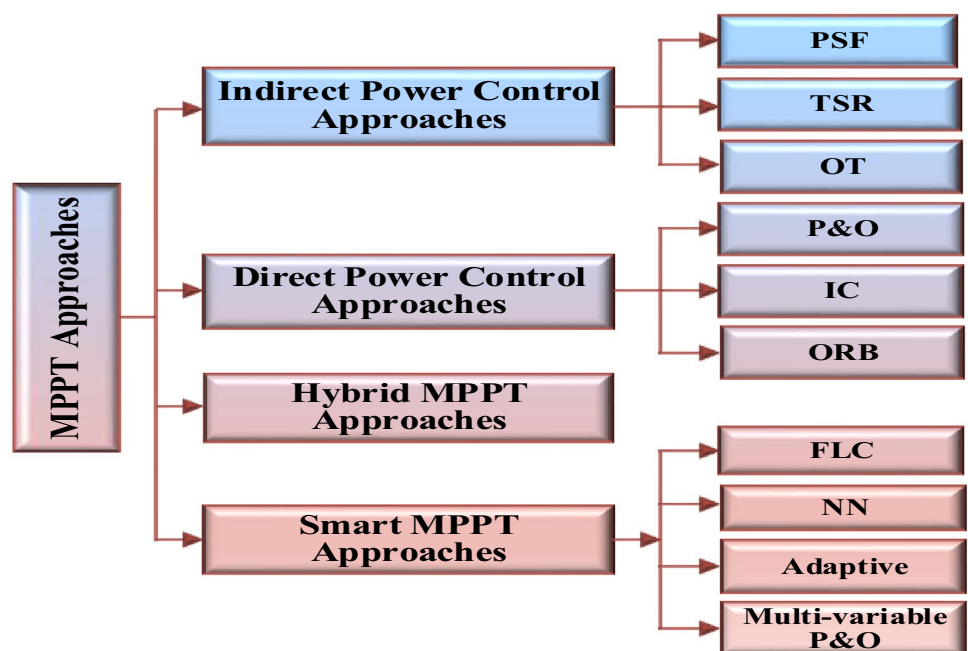


Table 6 The comparison between several MPPT algorithms

MPPT algo-rithm	Convergence speed	Complexity	Prior knowl-edge	Oscillations at MPP	Wind speed sensor	Memory require-ment	Performance under varying wind speeds	Cost	Efficiency
Indirect power algorithms									
PSF	Fast	Simple	Required	No	Yes	Yes	Moderate	Moderate	Moderate
TSR	Fast	Simple	Not required	No	Yes	No	Moderate	Very High	Very High
OT	Fast	Simple	Required	No	No	No	Moderate	Moderate	Moderate High
Direct power algorithms									
P&O	Slow	Simple	Not required	Yes	No	No	Moderate	Moderate	Low
IC	Slow	Simple	Not required	Yes	No	No	Moderate	Moderate	Low
ORB	Medium	Simple	Not required	No	No	Yes	Moderate	High	Moderate
Hybrid algo-rithms	Fast	Medium	Not required	Depends	No	No	Good	Moderate	Very High
Smart algorithms									
FLC	Medium	High	Required	No	Depends	Yes	Very Good	High	High
NN	Medium	High	Required	No	Depends	Yes	Very Good	High	High
Adaptive	Medium	High	Required	No	Depends	Yes	Very Good	High	High
Multi-vari-ble P&O	Slow	High	Not required	Yes	No	No	Good	Low	Moderate

ing the optimal power and its corresponding wind turbine speed. The main drawback of employing this algorithm is that it necessitates the use of several sensors to obtain previous knowledge about reference power. However, it is robust and cost-effective approach [112–114].

6.2.1.2 Tip Speed Ratio Algorithm It is necessary to maintain the tip speed ratio at an optimum value, where the extracted power is maximized by adjusting the rotational speed of generator. It needs to measure the speeds of wind and generator to get the optimal tip speed ratio (λ_{opt}) of turbine for extracting possible maximum power. The optimal rotational speed is determined from Eq. (2) as:

$$\omega_m^{\text{opt}} = \frac{\lambda_{\text{opt}} V_W}{R} \quad (3)$$

The block diagram of tip speed ratio (TSR) algorithm for WECS is shown in Fig. 14b. There are two primary ways to obtain wind speed; wind speed estimation or mechanical sensors. For mechanical sensors method, numerous anemometers are mounted around the swept area of wind turbine to measure the wind speed. These anemometers have low accuracy which leads to a reduction in the total efficiency and reliability with high maintenance and initial costs. While for wind speed estimation method, the operating wind speed can be calculated by accurate wind speed estimation algorithms to overcome the disadvantages of anemometer [112, 114].

6.2.1.3 Optimal Torque Algorithm The generator torque is regulated in this method to get optimum reference torque curve based on the wind turbine peak power at a particular wind speed. The expression of mechanical optimal torque for the turbine if the rotor is rotating at optimal tip speed ratio and power coefficient is:

$$T_m^{\text{opt}} = \frac{1}{2} \rho \pi R^5 \frac{C_p^{\text{opt}}}{\lambda_{\text{opt}}^3} \omega_m^2 = K_{\text{opt}} \omega_m^2 \quad (4)$$

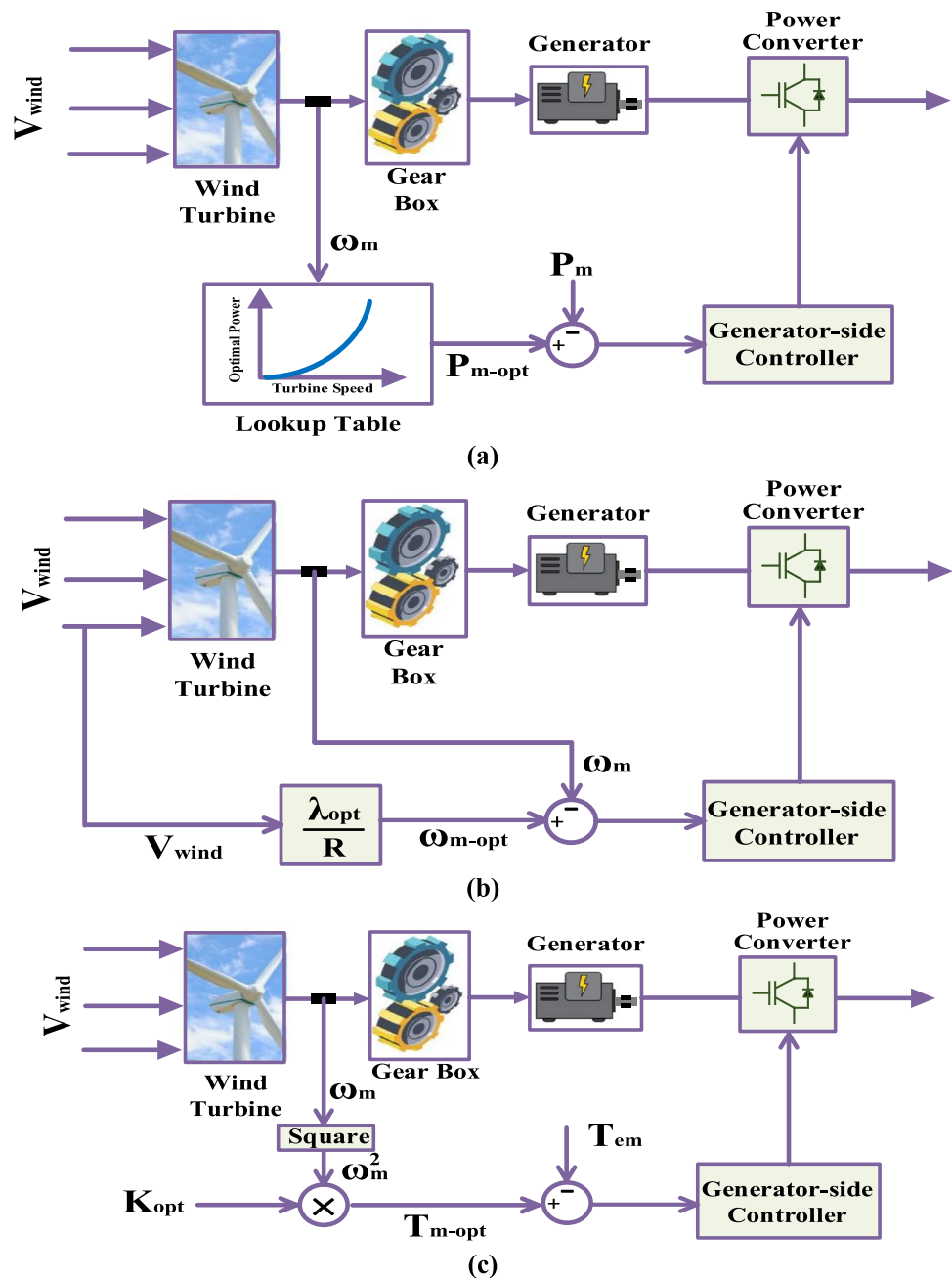
The block diagram of optimal torque (OT) MPPT algorithm for WECS is illustrated in Fig. 14c.

This algorithm is widely used in WECSs because of its advantages such as fast tracking with high efficiency and simplicity. But this algorithm needs prior information about the mechanical parameters of turbine as well as weather conditions [112–115].

6.2.2 Direct Power Controller Based MPPT Algorithm

6.2.2.1 Perturb and Observe Algorithm Perturb and observe (P&O) method is widely employed in WECS for maximum power extraction by searching for the MPP. It depends on disturbing a control variable in some step size and detecting the resulting variations in the objective function till the slope becomes zero. Therefore, it does not require anemometers and prior knowledge about wind turbine parameters [112–114]. This method is slow and detects wrong direction to reach the MPP during fast wind variations. The modi-

Fig. 14 The block diagram for
a PSF MPPT, **b** TSR MPPT, **c**
 OT MPPT



fied and adaptive step P&O algorithms efficiently solve this problem. These modified methods achieve an appropriate balance of control efficiency and tracking speed, as well as solving incorrect directionality problem of P&O algorithm under changing wind conditions. The direction and size of variable step for the next perturbation are intelligently decided by observation of the distance between the optimal curve and operating point. The block diagram of P&O MPPT algorithm for WECS is shown in Fig. 15a [112–114].

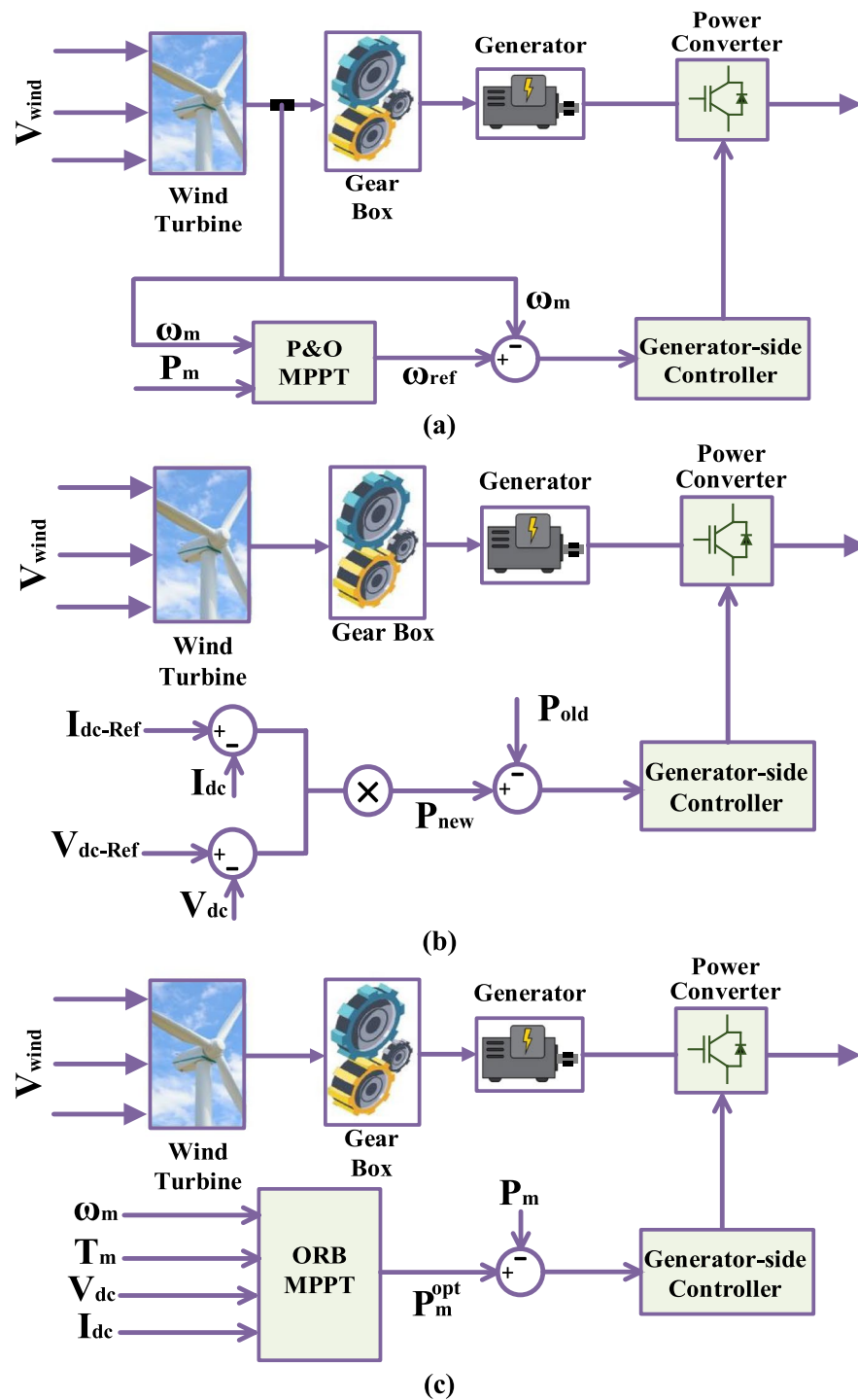
6.2.2.2 Incremental Conductance Algorithm This method does not need any sensors and generator and turbine param-

eters, resulting in improving system reliability and lowering costs. The output power of turbine can be expressed as a function of DC-link voltage where, there is an optimum DC-link voltage that maximizes the generator output power. Because power equals the product of voltage (V_{dc}) and current (I_{dc}), the slope can be calculated as:

$$\frac{dP_{dc}}{dV_{dc}} = \frac{d(V_{dc}I_{dc})}{dV_{dc}} = I_{dc} + V_{dc} \frac{dI_{dc}}{dV_{dc}} \quad (5)$$

For MPP, the expansion of this derivative is as follows:

Fig. 15 The block diagram for **a** P&O MPPT, **b** IC MPPT, **c** ORB MPPT smart MPPT algorithms



$$\text{Slope} = \frac{I_{dc}}{V_{dc}} + \frac{dI_{dc}}{dV_{dc}} \quad (6)$$

The basic concept is that the tangent slope of power–voltage characteristic must be zero at MPP, negative on the right side, and positive on the left side of MPP. Therefore, this method can tackle MPP by looking for rectifier output power rather than the observation of weather-dependent parameters

such as wind direction and wind speed. The block diagram of incremental conductance (IC) MPPT algorithm for WECS is illustrated in Fig. 15b. The performance of the IC MPPT algorithm can be enhanced by employing automatically adjusting the variable step size for V_{dc} changes. These modified IC algorithms provide convergence speed improvement and decrease the power variations around MPP [112–114].

6.2.2.3 Optimal Relation-Based Algorithm This MPPT method is based on the optimal relationships between variables such as turbine output power, wind speed, converter power, and DC current and voltage of converter, etc. The benefit of this method is not requiring a speed sensor or the use of a lookup table. It works on the basis of a previously established power versus DC current curve where, the MPP can be tracked at all times by working on the optimum current curve. The DC current for a certain wind speed must not exceed the curve of maximum current in order to maintain generating. The approximate relation between the optimal DC current (I_{dc}^{opt}) and the optimal DC voltage (V_{dc}^{opt}) when the system is at MPP is as follows:

$$K = \frac{I_{dc}^{opt}}{V_{dc}^{opt2}} \quad (7)$$

It is obvious from Eqs. (4) and (5) that if k is known by the control action, it is possible to determine the optimal curve and effectively apply the optimal relation based (ORB) MPPT method. The block diagram of ORB MPPT algorithm for WECS is shown in Fig. 15c [112–114].

6.2.3 Hybrid MPPT Algorithms

The hybridization of two or more MPPT methods is an effective and simple technique for overcoming the disadvantages of the classic MPPT algorithms. The ORB MPPT technique is adopted with self-tuning capability by employing the traditional P&O method. There are two modes of operation for this method. The first mode is using initially the traditional constant step P&O MPPT technique for searching for MPP at any wind speed. Then, the unknown parameter (k) in the ORB method may be easily computed once the MPP has been identified and the current and voltage are measured. The optimum current curve can be calculated after the k is obtained. The hybrid MPPT method then changes to the second mode, which can track the MPP using the ORB method equation. The PSF, OT and IC techniques can also be used with the P&O method to solve the shortcomings related to P&O algorithm, such as incorrect directionality during fast wind speed changes [112–114, 116, 117]. Many alternative MPPT methods are used by employing intelligent controllers, like the neural network (NN), fuzzy logic controller (FLC), and other soft computing techniques. Additionally, other MPPT algorithms use meta-heuristic optimization techniques to avoid the limitations of traditional MPPT algorithms. Currently, soft computing methods are the popular choice for WECSs since they do not necessitate any mathematical parameters for WECS. But

its disadvantage is the inaccuracy of its optimal response to the control variables. Therefore, several adaptive methods are developed to deal with the uncertainty of system for optimal circumstances without the need for mechanical sensors. Furthermore, the multivariable P&O MPPT algorithm is more effective for MPP tracking in WECSs [112–114].

Various control methods have been presented for MPPT applications that employ FLC either alone or in combination with other approaches. The major advantages of this controller are that the controller parameters can be varied relatively fast in response to system dynamic changes without parameter estimation, less oscillations around the MPP, and strong performance when changing the climatic conditions. However, their advantages are extremely dependent on the ability of user to select the appropriate surface error, membership function levels, and selection of the rule-base layer with significant memory space requirement. The NN algorithms have been significantly expanded for the MPPT applications where, they are used in several research studies to improve their proposed MPPT methods. In general, the NN structure has three layers: input, hidden, and output layers, with variable nodes number depending on the user judgement.

The optimum state of various control variables, as well as the inputs and suitable weights of layers are regulated by NN algorithms. Terminal voltage, pitch angle, wind speed, rotor speed, output torque, or any combination of these variables can be used as input variables. The output is commonly a reference signal such as reference rotor speed, power, torque, etc. which is employed to drive power electronic circuit to force WECS to operate at the MPP [112–114].

The adaptive MPPT technique, on the other hand, can measure uncertain system parameters and gives an accurate optimum response. It is dependent on a vector control scheme that transitions between multi-operational modes based on the detection of changeable wind speed. This approach is especially adapted to changes in climate using these specially-designed modes, resulting in significantly improving the performance of tracking [112–114, 118]. The goal of multivariable P&O MPPT technique is maximization the produced output power of wind farm while lowering the need for sensors and control units. Such this technique makes wind power plants economically viable since the extracted power is the same as that of conventional wind farms with decreasing required components number. The P&O MPPT method extension is adopted for optimizing the currents of all generators existing in the wind farm for maximization of the output power of wind farm. Using the P&O concept, the multivariable P&O MPPT method is developed for optimization of the operating point for each generator. The current of the first

generator is perturbed in this approach, while the aggregated output power rises. The procedure is carried out frequently to cover all the available generators in the wind farm [112–114].

6.2.4 Recent MPPT Algorithms

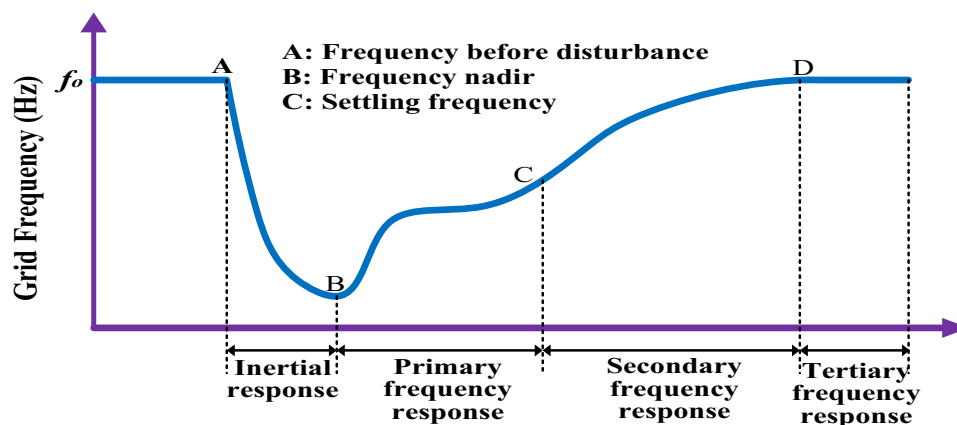
This subsection introduces the overall view of the recent research on the improvement of MPPT algorithms. In [119], Yang et al. presented a novel passivity-based sliding mode control approach for achieving MPPT for PMSG based WECS. This study was concluded that the proposed control achieves MPPT quickly with less overshoot under gradual and random wind speed changes. In [120], Zhang et al. introduced an improved MPPT control scheme which is based on decreasing torque gain control method. It dynamically adjusts the gain coefficient to maximize energy capture in low wind locations. In [121], Chen et al. proposed a robust MPPT control approach for a grid connected PMSG based WECS by using perturbation and observation based nonlinear adaptive controller for enhancing the efficiency of energy conversion. Parameter uncertainties, system nonlinearities, and external disturbances are represented as lumped perturbation term that is estimated by high-gain perturbation observer in the proposed approach. In [122], Chen et al. presented a novel robust data-driven approximate optimal MPPT control method for WECS by using the adaptive dynamic programming technique combined with a recurrent neural network. A recurrent neural network first constructs a data-driven model to recreate the dynamics of wind power system by employing available input–output data. Then, adaptive dynamic programming method is used to create the approximation optimum tracking controller, that comprises of the optimal feedback and steady-state controllers. In [123], Youssef et al. introduced an intelligent self-adaptive P&O and fast-hybrid P&O techniques for WECSs for enhancing the dynamic performance and overcoming the limitations of current P&O methods. The proposed methods

depend on the estimation of the optimal relationship between rotor speed and power to accurately adapt step sizes per-perturbation under various environmental conditions. They decrease MPP search area to 10% of the optimum power and speed curve with on need for prior knowledge of perturbation step-sizes and without dividing the curve into modular operation sectors. In [124], González-Hernández et al. proposed a novel MPPT PI discrete reverse-acting controller based on TSR method for WECS for the extraction of peak power from wind, maintaining a basic structure which can be easily implemented in any reasonably priced numerical digital processor. In [125], Wang and Bo presented adaptive fixed-time control approach to the MPPT algorithm for DFIG based WECS. The fixed time observer was employed for the estimation of aerodynamic torque without the need to wind speed sensors. Next, without knowing the upper limits of the system uncertainties in advance, an adaptive fixed-time controller was presented for implementing optimum rotation speed tracking for MPPT. In [126], Buduma et al. introduced a robust new control and Kalman filter for a grid-connected WECS for controlling the reactive and real power. The proposed Kalman filter based MPPT was developed for high gain boost converter that improves the performance of maximum power tracking.

6.3 Frequency Controller

Nowadays, the power networks inertia has been decreased as wind energy has become more widely integrated into the current power grid. In the event of a substantial frequency disturbance, the capability of system frequency regulation tends to deteriorate. Therefore, wind turbines can participate in controlling the network frequency to maintain the network frequency stability. Inertia and droop controls are commonly used for frequency control using derivative and proportional control. Figure 16 shows the response of grid frequency under power disturbances [127–130]. To ensure that variable speed wind turbines perform the function of

Fig. 16 The grid frequency response under power disturbances



frequency regulation correctly in accordance with certain requirements of frequency regulation, supplemental controllers must be developed and integrated into both the existing pitch angle control loop and the converter power control loop for manipulation of the corresponding reference set points of real power, torque, or the speed of rotor. A variable speed wind turbines must be operated in a suboptimal mode through deloaded control to participate in secondary, primary, and tertiary frequency control. This guarantees that a specific amount of spinning reserve margin is available at all times to provide additional real power in the case of frequency contingency. Currently, primary reserve is generally established by balance, delta, or fixed reserve control. A constant percentage of rated power is reserved through balance control while, delta control reserves a certain percentage of the available peak real power. For fixed reserve control, a constant amount of real power is reserved. The level of the reserve margin depends on the magnitude of the prevailing wind speed, the wind speed prediction accuracy, and the upper permissible limit of the variable speed wind turbine rotor speed [125–128].

The primary frequency control of WECS is similar to that of a traditional generator, which normally within a few tens of seconds is activated and continued for up to 15 min as soon as the deviation of grid frequency exceeds the permitted threshold. For emulating the conventional governor response, droop control is used in variable speed wind turbine for correlating the changes in the frequency of grid with corresponding changes in the output of real power via the controller of power converter. It adjusts the output of power in response to large grid frequency changes. The relationship between frequency deviation and real power change can be expressed as [127–130]:

$$\Delta P_{dr} = K_{dr}(f - f_o) \quad (8)$$

$$R = \frac{\Delta P_{dr}}{\Delta f} = -\frac{1}{K_{dr}} \quad (9)$$

where f_o is the rated frequency and the parameter (K_{dr}) is the speed droop inverse (R) which is defined as follows:

Secondary frequency control is employed both under normal operating conditions and during emergency frequency occurrences. It begins within seconds and lasts for multiple tens of minutes. This control is a continuous response on the minute timescale for allocating the change of load among individual wind turbines with the goal of keeping the tie line power flow and system frequency deviations at zero. The secondary frequency controller based on the wind farm supervisory control system fully uses secondary frequency reserve to adjust the power flow and follow up the commands from the system operator [127–130]. Tertiary frequency control has longer decision time step from minutes

to hours in comparison to the other frequency controls. It takes action after secondary control. It includes dispatching procedures from the operator of system to accomplish power reserve and recovery for tertiary frequency control for WECS in order to provide economic dispatch, unit commitment, and optimal power flow in response to market signals or other system requirements [127–130]. The use of combination of different frequency controllers can improve overall frequency regulation performance of power system and reduce the stress of frequency regulation in traditional units. The coordinated frequency control process can be described as follows. Once occurring frequency drop, the inertial response plays a vital role in the reduction of frequency change rate and the increase of frequency nadir point. When the dead-band condition for a certain delay time is satisfied, primary frequency response is activated to further improve the frequency regulation by droop control till the frequency reaches to secondary steady state. Then, secondary frequency control is employed to accomplish zero frequency error regulation by using PI controller [127–130].

6.3.1 Recent Frequency Control Methods

This subsection provides an overview of recent research on frequency regulation control methods, which are discussed below:

In [131], Zhang et al. presented a frequency regulation control approach by wind power based on limited over-speed de-loading curve partitioning. The range of over-speed control was quantified, and the restricted over-speed de-loading curve was developed and fitted. This technique improves the usage of over-speed de-loading while also improving the wind turbine frequency regulating capability. This study was concluded that the proposed approach can efficiently enhance the frequency stability and system inertia and enhance the response of frequency under complex operating conditions for wind power systems. In [132], Peng et al. introduced a novel fuzzy theory based coordinated control approach to enhance the short-term frequency response capability of a DFIG-ESS. Four wind speed zones were categorized because of making the short-term frequency response capability of the DFIG-energy storage system adaptive for different conditions of wind speeds. Two types of FLCs were designed and used in the proposed control where its inputs are DFIG operation state and frequency deviation and the its output are DFIG participation coefficient for short-term frequency support and the power support from the combined system. The results of simulation were demonstrated that the proposed control can efficiently enhance the short-term frequency response capability of the DFIG-ESS under different wind speed conditions. In [133], Hu and Wu proposed a new linear-gain droop control loop for a DFIG based wind farms to approximate the capability

of frequency control. The gain of droop is proportional to the rotor speeds of wind turbine. The proposed linear droop gain can provide a satisfactory approximation to the quadratic droop gain by selecting the suitable coefficients of the linear function. The results of simulation demonstrated that the proposed scheme is able to closely approaching the kinetic energy-based droop control loop. In [134], Li et al. presented a de-loading control approach which combines pitch control and over speeding of wind turbines operating at varying wind speeds. The proposed control approach can regulate wind turbines static frequency difference coefficient, that is dependent on the proportion of variable-speed wind turbines running at high speeds of wind. Additionally, it supports frequency control of the grid under power limitations, and appropriately supports frequency regulation for wind turbines operating at various wind speeds. This study was concluded that the control approach can enhance the primary frequency control performance. In [135], Verij Kazemi et al. introduced a new approach for adaptive tuning of gains for second-order derivative terms (PDD2) controller in DFIG for frequency control. The proposed approach is based only on control system output and input data, which aids in overcoming system modelling problems. A parabolic sliding-mode filter was also employed to eliminate measured frequency noise. The results of simulation were indicated that the efficacy and robustness of the proposed control in the steady state and transient for supporting grid frequency in wind turbine. It enhanced the frequency nadir by minimum 18% as compared to the conventional frequency control approaches. In [136], Xu et al. proposed an enhanced droop control method for a DFIG which can reduce the maximum frequency change rate and frequency nadir during the frequency regulation. The proposed droop control coefficient is a linear function of the frequency change rate. So, it can regulate the control coefficient according to the frequency change rate. This study was concluded that the proposed approach is more successful in improving the maximum rate of frequency change and frequency nadir under varied disturbance sizes, even with fluctuating wind speed. In [137], Yang et al. presented an improved frequency control strategy using dynamic droop coefficients of type 3 WECS that effectively mitigates system frequency variations. The results of simulation were indicated that the proposed frequency control method is useful to enhance the capability of frequency regulation under several patterns of wind speed and the penetration levels especially for power systems with highly wind generation. In [138], Chen et al. introduced a technique for the estimation of the virtual moment of inertia given by a DFIG-based wind farm with additional frequency control. Routh approximation was used to derive the explicit formulation of the virtual moment of inertia for grid connected DFIG-based WECS. The proposed expression of the virtual moment of inertia is validated by numerical results

tested by a modified IEEE 30-bus system and DFIG grid-connected system. It was concluded that the proposed estimation method is conventional for power systems with a lot of wind energy. In [139], Yang et al. proposed an adaptive droop control approach with smooth rotor speed recovery capability for type 3 WECS for adjusting the time to meet MPPT curve, enhancing the frequency nadir, and minimizing the second drop in frequency. The first purpose was achieved by making the ratio of the proposed control output to MPPT output reference proportionate to rotor speed, which is mostly determined by wind speed. While, the two objectives were realized by the use of the proposed adaptive droop gain that reduces with time. It was concluded that the proposed control approach prevents DFIGs from stalling, improves the frequency nadir, and reduces second drop in frequency.

7 AI Techniques in Grid-Connected WECSs

The following section provides an overview of artificial intelligence (AI) techniques in grid-connected WECSs. Optimum control of large-scale wind farms and their integration into power grids is a vital challenge for providing reliable, effective, and secure electrical energy. As a result, AI techniques and soft computing methods can be used with conventional control approaches to effectively enhance the system performance in WECSs. Many soft computing-based optimization techniques like artificial NN, FLC, genetic algorithm, particle swarm optimization, etc. have been proposed for WECS. It is verified by many researchers that AI techniques can be efficiently employed in controller parameter tuning, MPPT, and LVRT methods for grid-connected WECSs. The recent AI techniques in grid-connected WECSs are tabulated in Table 7 [140–142].

7.1 AI Techniques for Controller Parameter Tuning

7.1.1 DFIG Based WECS

In [143, 144], Elkholy et al. presented an effective controller of wind turbine DFIG based on teaching learning-based optimization (TLBO) algorithm and artificial neural networks to have MPPT and optimal active and reactive power control. In [145], Yang et al. proposed a novel grouped grey wolf optimization for obtaining the optimum PI parameters of generator-side controller for DFIG-based WECS for achieving a MPPT together with improving FRT capability. The results of simulation showed that the proposed method can capture the maximum wind energy and efficiently restore the system after occurring a fault at the electrical grid. In [146], Madan and Kumar introduced an optimized LVRT control by

Table 7 Recent AI techniques in grid connected based WECSs

Refs.	Algorithm adopted	Year	Description	Core objectives
[143]	Teaching learning-based optimization algorithm	2016	• Implemented for DFIG based WECS	• To present an effective and intelligent controller to extract the maximum power from wind turbine based on rotor voltage control
[144]	Teaching learning-based optimization algorithm	2017	• Implemented for DFIG based WECS	• To have optimal active and reactive power controller over a wide range of wind speed based on AI techniques
[145]	Grouped grey wolf optimizer	2017	• Implemented for DFIG based WECS	• For achieving MPPT together with an improved LVRT
[146]	Grey wolf optimization with fuzzified error	2019	• Implemented for DFIG based WECS	• For improving LVRT
[147]	Bacteria foraging optimization	2020	• Implemented for DFIG based WECS	• To adjust the rotor currents and the DC-link voltage
[148]	Thermal exchange optimization	2020	• Implemented for DFIG based WECS	• To obtain optimal gains of PI in rotor-side and grid-side controllers
[149]	Salp swarm algorithm	2021	• Implemented for BDFIG based WECS	• To adjust the torque, speed, current, reactive, and active power
[150]	Particle swarm, gray wolf, moth-flame, and multi-verse optimizers	2021	• Implemented for DFIG based WECS	• To tune the FLC parameters for both grid-side and generator-side controllers to enhance the dynamic operation
[87]	Modified elephant herding technique and random forest technique	2021	• Implemented for DFIG based WECS	• For enhancing the LVRT capability
[151]	Particle swarm optimization algorithm	2021	• Implemented for DFIG based WECS	• For obtaining damped behavior and minor peaks in the mechanical and electrical variables during occurrence electrical faults
[152]	Water cycle technique	2021	• Implemented for DFIG based WECS	• For parameters tuning of PI controllers in vector control scheme
[153]	Quantum parallel multi-layer Monte Carlo optimization technique	2021	• Implemented for DFIG based WECS	• For optimizing the rotor-side controller parameters based on PI of DFIG to achieve MPPT and enhance the efficiency of generation
[154]	Democratic joint operations	2018	• Implemented for PMSG based WECS	• To obtain the optimal gain parameters of PID controllers to achieve the optimal MPPT
[155]	Grey wolf optimizer	2018	• Implemented for PMSG based WECS	• To tune the optimal PI controllers gains to improve MPPT, LVRT capability, and the steady-state operation
[97]	Cascaded ANFIS-GA	2018	• Implemented for PMSG based WECS	• To enhance the system dynamic stability
[156]	Whale optimization algorithm-based Sugeno FLC	2019	• Implemented for PMSG based WECS	• For improving the FRT capability
[157]	Grasshopper optimization algorithm	2020	• Implemented for PMSG based WECS	• To enhance the transient stability performances
[158]	Transient search optimizer	2020	• Implemented for PMSG based WECS	• To enhance the transient stability performances
[159]	Hybrid cuckoo search algorithm and grey wolf technique	2021	• Implemented for PMSG based WECS	• For enhancing the system dynamic stability and extracting maximum power from PMSG at unity PF
[160]	Sine cosine algorithm	2021	• Implemented for PMSG based WECS	• To extract maximum wind energy and improve the WECS performance under dynamic operation conditions

using the grey wolf optimization technique with fuzzified error algorithm. This paper compared the proposed optimized method with other traditional optimization

algorithms. It was concluded that the proposed technique is more effective in simulating the LVRT system as compared to other traditional methods. In [147], Bakir et al.

introduced a bacteria foraging optimization technique to tune PI controllers parameters in a grid connected DFIG-based WECS. The effectiveness of proposed optimization strategy was assessed using a variety of performance criteria based on tracking errors. In [148], Alhato and Bouallègue introduced a thermal exchange optimization algorithm for obtaining the optimal gains of PI controllers for a DFIG-based WECS. The tuning of the gains of PI controllers was investigated as an optimization problem against non-smooth and nonlinear operational limitations. In [149], Memon et al. presented a salp swarm algorithm-based vector control to enhance the transient response of a BDFIG-based WECS. The proposed method was used for the tuning of PI controller to optimally adjust torque, current, speed, reactive, and active power of the system in the case of sharp variations in the load and wind speed. In [150], Nasef et al. introduced an optimum tuning of FLC for a 1.5 MW grid connected DFIG-based WECS by using intelligent methods like the gray wolf, particle swarm optimizer, multi-verse optimizer, and moth-flame optimizer for both grid and generator side controllers. A multi-objective optimization was proposed with the aim of reducing the steady-state errors of these controllers to enhance the transient operation of the system under variable wind speed conditions. It was concluded that moth-flame optimizer-FLC has the best performance as compared to other various optimization techniques regarding the convergence speed, controller step response and best fitness function value. In [87], Manohar et al. presented a hybrid algorithm which combines both modified elephant herding technique and random forest technique for improving LVRT capability of the DFIG based WECS. In the proposed method, several parameters related to LVRT like, current, voltage, and active and reactive powers were used as objective function which was solved by proposed method. In [151], B. Aguilar et al. proposed a particle swarm optimization for the tuning of the parameters of PI controllers for obtaining damped behaviour and minor peaks in the mechanical and electrical variables during the occurrence of electrical faults. The results of simulation showed that the proposed method is an effective for continuous and nonlinear problems, accomplishing an acceptable tuning of the PI controller parameters. In [152], Bakir et al. introduced a water cycle optimization technique for tuning six control PI parameters by reducing a cost function carried out using the tracking errors in rotor-side and grid-side controllers in a grid connected DFIG-based WECS. In [153], Han et al. presented a novel quantum parallel multi-layer Monte Carlo optimization for optimizing the rotor-side controller parameters based on PI of DFIG to achieve MPPT and enhance the efficiency of generation. The proposed method was used to obtain the best solutions in a precise small search space.

This study was concluded that the proposed method can provide the result of optimization objective function of the rotor-side controller with at least 0.51% lower than other techniques and improve the average power coefficient of DFIG based WECS by at least 0.0028%.

7.1.2 PMSG Based WECS

In [154], Yang et al. presented a novel democratic joint operations optimization technique to obtain the optimal PID controllers parameters of PMSG-based WECS for achieving the optimal MPPT under several wind speed profiles. In [155], Qais et al. proposed a grey wolf optimization for tuning the optimal gain of eight PI controllers used in the control of generator-side and grid-side converters to improve the MPPT, LVRT capability and the steady-state operation by obtaining of the minimum of the summation of integral-squared error of produced real power error, DC voltage error, and terminal voltage errors of the grid and PMSG. In [97], Soliman et al. introduced a novel cascaded adaptive neuro-fuzzy inference system-genetic algorithm based control strategy to improve the performance of the PMSG based grid-connected WECS. The proposed control strategy was employed to control both of the grid-side and generator-side converters where, GA algorithm was proposed to adapt the ANFIS parameters. In [156], Qais et al. presented an optimal design of multi-sugeno FLCs based on the whale optimization technique to enhance the FRT capability of grid connected PMSG based WECS in the event of unbalanced and balanced grid fault conditions. The whale optimization was used to design the rules of control and the Gaussian memberships of 8 sugeno FLCs, simultaneously, by reducing the high-dimensional multi-objective objective function. In [157], Amin et al. proposed a grasshopper optimization technique based PI controller for the improvement of transient stability performance of grid connected PMSG based WECS. The proposed method was used in the process of optimization, where the integration of square error criterion was considered as objective function. In [158], Amin et al. presented a novel transient search optimizer to optimally design the PI controllers that are used in generator-side and grid-side controllers for improving LVRT capability for grid connected PMSG-based WECS. The transient search optimizer was coded by using the Fortran language within the PSCAD software. The effectiveness of proposed method was compared with that accomplished by using a recent grey wolf optimization under various grid disturbances. In [159], Amin et al. proposed a new hybrid cuckoo search technique and grey wolf optimization-based on control method for improving the dynamic stability of grid connected PMSG-based WECS. The grid-side and generator-side converters were controlled by using an optimal PI controller for the extraction of maximum power from the PMSG at unity PF.

In [160], Shutari et al. proposed a novel application of the sine cosine technique which aims to obtain the finest PI gains values leading to the extraction of maximum wind power and improving the WECS performance under random wind speed changes and severe grid disturbance. The proposed method was employed to minimize the sum of the integral squared error of 12 PI controllers error inputs in the control methods simultaneously.

8 Conclusion

This paper introduced a comprehensive review of grid-interfaced WECSs with main focus on electrical view. An overview of different topics concerning the WECS configurations, electrical generators and various power converter topologies are investigated. Furthermore, the international grid codes of frequency, power factor, power quality, harmonics and LVRT capability requirements were discussed. The major controller approaches for grid and generator converters such as MPPT, frequency controller, generator-side and grid-side converter controllers were analyzed. Additionally, the recent AI techniques which employed in the controllers of grid-connected WECSs were introduced.

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Research Involving Human and/or Animal Participants The study did not involve humans or animals.

Informed Consent The study did not involve humans.

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